

Improving the potential of fifth-generation district heating and cooling networks through robust design and operational optimization under future energy market and demand uncertainties

Afraz Mehmood Chaudhry^{a,b,*}, Ghader Ghorbaniasl^{a,b}, Jonathan Hachez^{a,b},
Stanislav Chicherin^{a,b}, Svend Bram^{a,b}

^a Thermo and Fluid Dynamics (FLOW), Faculty of Engineering, Vrije Universiteit Brussel (VUB), Pleinlaan 2, Brussels, 1050, Belgium

^b Brussels Institute for Thermal-Fluid Systems and Clean Energy (BRITE), Vrije Universiteit Brussel (VUB) and Université Libre de Bruxelles (ULB), Brussels, 1050, Belgium

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ABSTRACT

Network investments and projected energy prices greatly impact the financial viability and environmental impact of fifth-generation district heating and cooling (5GDHC) networks. Compared to the previous generation, the upfront investment costs in 5GDHC networks are relatively high due to the decentralized energy demand and supply of the participating prosumers. The transition to 5GDHC is also hindered by uncertain energy markets and the fluctuating energy demands of the prosumers. It makes investors hesitant to commit, even for promising business cases. In this work, a framework is presented to assess the economic and environmental performance of a 5GDHC business case, based on a multi-objective robust optimization algorithm (MO-RDO), to identify the optimal trade-offs between environmental and economic designs. The proposed methodology consists of four steps. In the first two steps, a specific business case is modeled, and techno-economic and environmental performances are analyzed against uncertain energy markets. In the third step, the techno-economic and environmental indicator responses are used to develop a data-driven machine-learning model. This model offers better computational efficiency than a high-fidelity nonlinear model and assesses the most influential parameters driving economic and environmental performance via a global sensitivity analysis. This aids in refining the multi-objective robust design optimization (MO-RDO) framework by pinpointing key design variables and objectives amid data uncertainties. Additionally, the formulated MO-RDO provides a mix of environmentally and economically optimal network designs and operational parameters and highlights the trade-offs between them. The designs and trade-offs are evaluated using financial metrics like net present value (NPV), return on investment, and discounted payback period. These metrics are calculated over the project life, considering the initial investment and net savings. For the considered case, the environmental design has a 10% higher investment cost compared to the economically optimal solution but reduces total emissions (TE) by 13% and improves operational cost savings, which is crucial for better financial indicators. The proposed framework aligns with the EU transition plan to incorporate waste energy sources and decarbonization paths, estimating only a 3% increase in NPV as a risk for the environmentally optimal solution on a financial scale.

1. Introduction

1.1. Context

Over half the world's population resides in urban areas, a figure expected to rise 1.5 times by mid-century due to population growth and urbanization. Urban areas currently account for over 70 % of final energy use and greenhouse gas emissions [1,2]. Thus, urban energy

planning become an important research topic for sustainable development of cities and societies and to mitigate greenhouse gas emissions. Although there is no universal solution exists in answering sustainable development, however, district heating and cooling has been recommended as a robust solution [3,4].

Fifth-generation district heating and cooling (5GDHC) is the latest district heating and cooling concept that operates at ultra-low network temperatures [5–8]. The main idea is to lower the network temperature

* Corresponding author at: Thermo and Fluid Dynamics (FLOW), Faculty of Engineering, Vrije Universiteit Brussel (VUB), Pleinlaan 2, Brussels, 1050, Belgium.
E-mail address: afraz.mehmood.chaudhry@vub.be (A.M. Chaudhry).

Nomenclature

χ	operational cost savings, EUR/y	Q	thermal energy, kWh
AC	annualized cost, EUR/y	R	building reference profiles, kW
Δp	pressure drop, Pa	r	discount rate, %
η	efficiency, %	S	slope line of heat pump, rC/rC
$\dot{m}$	mass flow rate, kg/s	W	electric energy, kWh
ω	vector of uncertainties, –	X	random inputs
$\dot{Q}$	thermal power, kW	Y	random outputs
ρ	density, kg/m ³	5GDHC	fifth-generation district heating and cooling
$e_{el/gas}$	specific emission of electricity or gas, kgCO ₂ eq/kWh	ASHP	air source heat pump
CAPEX _c	centralized capital expenditures, EUR/y	ATES	aquifer type energy storage
CAPEX _d	decentralized capital expenditures, EUR/y	BTES	borehole type energy storage
CAPEX _o	overall capital expenditures, EUR/y	CAPEX	capital expenditures
CAPEX _r	reference capital expenditures, EUR/y	COP	coefficient of performance
CR	cost ratio, %	DPBP	discounted payback period
EM _c	emissions centralized, kgCO ₂ eq/y	GCHP	ground coupled heat pump
EM _d	emissions decentralized, kgCO ₂ eq/y	GHE	ground heat exchange
EPC	energy performance certificate, MWh/m ²	GIS	graphic information system
OPEX _c	centralized operational expenditures, EUR/y	GSHP	ground source heat pump
OPEX _d	decentralized operational expenditures, EUR/y	HDSS	heat distribution system simulator
OPEX _o	overall operational expenditures, EUR/y	IRR	investment return rate
OPEX _r	reference operational expenditures, EUR/y	KPI	key performance index
SA	surface area of the building, m ²	ML	machine learning
SPBP	standard payback period, y	MO	multi objective
TE	total emissions, tonCO ₂ eq/y	NG	natural gas
$\dot{W}$	electric power, kW	NPV	net present value
c_{CO_2}	penalty of CO ₂ , EUR/kWh	OPEX	operational expenditures
$c_{el/gas}$	specific cost of electricity or gas, EUR/kWh	PCE	Polynomial chaos expansion
C_{EM}	cost of emissions, EUR/y	RDO	Robust design optimization
c_p	specific heat capacity, kJ/kg°C	SC	space cooling
D	diameter, m	SH	space heating
L	length, m		

to allow low-grade waste heat recovery and adopt the temperature levels as per user requirements in network substations with heat pumps [9]. The transition towards this system is limited due to significant challenges, both in research and implementation. A limited number of 5GDHC systems are operational and technology is relatively new and has a limited number of design guidelines [4]. In Gudmundsson et al. [10], economic comparison of 5GDHC systems is conducted with fourth generation district heating (4GDH). The centralized aspects of 4GDH offer clear economic benefits compared to 5GDHC as reported in [11]. Buffa et al. [9] also pointed out that 5GDHC substations are more expansive than those in previous DH substation [11–13]. Further, the operations and control design in a prosumer based 5GDHC network is also complicated [14]. Additionally, uncertainties in energy demands, energy prices, and equipment costs contribute to the challenges in network design and operational optimization as discussed in Egberts et al. [15], Neri et al. [16]. All these challenges, in turn, make the 5GDHC business case less attractive to potential investors [3].

1.2. Previous work

The development of 5GDHC networks has been explored in various research studies. [4] provides a critical analysis of recent advancements from the demonstration phase as well as the literature. Additionally, they emphasized cutting-edge approaches for the modeling and optimization of 5GDHC networks. Further, the research on the environmental and economic performance of the 5GDHC networks is also examined. This study's objectives are to clarify the meaning of 5GDHC networks and outline their benefits and drawbacks. According to [9], a study was carried out on 40 thermal networks in Europe that could supply both heating and cooling. Similarly, Volkova et al. [17] highlights the challenges of changing a high-temperature network grid to a low-

temperature one due to financial limitations. It explains that the time it takes to recover the investment made in this transition is longer than expected, according to a thorough analysis. Wack et al. [18] mentioned that the upfront investment is a crucial factor for the DH technology. Thus, in [18], a pipe penalization approach for the economic topology optimization of District Heating Networks is proposed, which yields the optimal net present value of the network.

In Calise et al. [19], a comparative analysis between 4GDHC and 5GDHC systems is conducted using a SWOT analysis and thermo-economic evaluation. Each network type employs separate rings to provide heating and cooling demands. Notably, the 4GDHC system operates at elevated network temperature levels, enabling the utilization of high-grade waste energy from power plants and CHPs. While both network types offer significant energy savings, their economic and financial performance is limited, with payback periods estimated at 16 years for 4GDHC and 20 years for 5GDHC. The research suggests that 4GDHC is optimal for residential districts, whereas 5GDHC is better suited for areas with simultaneous heating and cooling demands, offering greater profitability and risk mitigation as outlined in the SWOT analysis by [9]. Hennessy et al. [20] also highlighted that despite the decentralized consumption and production in 5GDHC networks leading to high energy costs associated with electricity prices, the Net Present Value (NPV) and Discounted Payback Period (DPBP) remain appealing under market conditions.

In Arabkoohsar et al. [21], an innovative and cost-effective configuration of smart energy buildings with energy networks is examined. The concept of smart energy buildings involves the integration of new technologies such as batteries, heat pumps, photovoltaic (PV) panels, photovoltaic thermal (PVT) systems, and photovoltaic thermal and cooling (PVTC) systems ([22,23]). Arabkoohsar et al. [21] developed a com-

prehensive techno-economic solution to determine the NPV and simple payback period (SPBP) for various configurations of these technologies when integrated with the network. The comparison of NPV is conducted over a project lifespan of 40 years. Among the different solutions analyzed, a configuration with PVTC systems and heat pumps demonstrated a higher NPV and an SPBP of less than 6.3 years.

In [24], the techno-economic performance of a 5GDHC system for a residential district located on Pantelleria Island is evaluated. The study examines the integration of wind turbines and PV panels for power production. Two separate rings within the 5GDHC network are considered, with fluid temperatures controlled by heat pumps attached to each ring and cooled by seawater. Calise et al. [24] that increasing the installed capacity of renewable energy plants does not significantly affect NPV, SPBP and DPBP. Due to high investment costs, the NPV is negative, and both SPBP and DPBP are 15 and 20 years, respectively. The SPBP stabilizes with the certain capacity increase of the renewable plant, as less electricity is sold to the grid.

In a related study [25], the integration of PV panels with a 5GDHC network is analyzed, and a thermo-economic analysis is conducted. The study investigates the decentralized generation of heating and cooling using heat pumps powered by installed capacities of PV panels. Although this configuration offers improved environmental benefits, the financial analysis reveals a high single project payback period (SPBP) of 33 years. The analysis reveals that PV panels only meet 30% of the electricity demand due to their intermittent nature. This suggests that integrating energy storage could alleviate the high SPBP, which is primarily driven by the substantial upfront costs associated with installing heat pumps and pipes. In [26], a hybrid approach is employed, combining physical models with a linear programming method to calculate performance metrics such as the levelized cost of energy and diversity factor in heating and cooling demands. Importantly, their linear program also incorporates investment and associated operational costs of the equipment to maximize the overall NPV of the project. An advanced approach to configuring 5GDHC networks by leveraging overlapping building demands in Europe is explored in [27]. The study employs Geographic Information System (GIS)-based methods and data sources to analyze various regions.

The literature discussed above primarily focused on performance analysis without considering uncertainties arising from energy markets and network participants' demands, as elaborated in [3,28]. In [3], over 10,000 scenarios were examined through Monte Carlo simulations and sensitivity analysis. Additionally, Coppitters et al. [29], De Meulenaere et al. [30] emphasized that analyzing energy systems using fixed values, generic ranges, or precise parameter distributions from the energy market often leads to sub-optimal designs that deviate significantly from reality. To address this issue, a more comprehensive, robust analysis incorporating input uncertainties through probability distributions is necessary to achieve robust designs.

In a recent study by Zhang et al. [3], a five-step framework was developed, integrating energy system optimization with stochastic simulations, uncertainty analysis, and sensitivity assessment. This framework was applied to a small 5GDHC network covering an area of 0.25 km². The performance index of cost-saving probability was chosen to assess the potential for constructing 5GDHC networks, revealing that area demand density, overlapping heating and cooling demands, and linear demand density are the most sensitive factors. Furthermore, the study highlighted the increasingly critical role of investment in Thermal Energy Storage (TES) with the integration of a large share of renewable energy. In [15], a systematic method is presented to optimize network design and operations amidst uncertain energy demands. The authors formulated an optimization cost function that determines the appropriate network configuration, specifically sizing the network pipes, in response to uncertain energy demands. Similarly, Akbari et al. [31] conducted a related study on designing a network for a local community under demand uncertainty, where they assessed the impact of uncertain demands on pipe lengths. In [32], a detailed thermo-hydraulic

simulation model is constructed using TRANSYS and Fluidit Heat. This model is utilized to predict economic performance under the influence of multiple uncertainties generated through Monte Carlo-based simulation methods. The study further explores the economic viability of solar-assisted renewable technologies with 5GDHC systems. The levelized cost of heating is selected as a performance indicator when assessing the impact of multiple uncertainties. In their study, Marx et al. [33] conducted a risk assessment regarding the economic viability of district heating (DH) networks in the context of uncertain energy and waste heat availability. Additionally, they incorporated individual heat supply systems into their analysis. They employed a Monte Carlo-based approach to assess the impact of uncertainty on both individual systems and network scenarios. Their simulation results revealed that individual systems are more significantly affected by uncertainties compared to the network scenario.

Abokersh et al. [34] developed a systematic framework to perform multi-objective optimization using a robust machine learning approach to inherent sustainable performance and design of solar-assisted district heating systems (SDHS). The framework first quantifies the uncertainty in the context of SDHS design using global sensitivity analysis followed by a heuristic optimization approach. The framework is applied across different urban community sizes (10, 25, 50, and 100 buildings) situated in Madrid. The findings highlight significant enhancements in economic and environmental benefits associated with SDHS deployment, particularly when considering the inclusion of seasonal storage tank (SST) construction properties in the optimization process. Abokersh et al. [34] employed a feed-forward neural network approach to construct a meta-model formulation [35]. In the development process of such models, a wide range of training points are introduced through data generation (also known as sampling) to verify the accuracy convergence and time consumption of the meta-models [36]. In [34], data generation is carried out by utilizing a detailed model in TRANSYS. Finally, a Bayesian optimization approach is utilized to enhance the model accuracy and conduct sensitivity analysis. Chaudhry et al. [37] devised a framework for generic thermodynamic modeling of 5GDHC to analyze the techno-economic performance of potential business cases. In their framework, a systematic strategy is developed to analyze the network that looks promising in the preliminary analysis of GIS via uncertainty quantification. Therefore, Chaudhry et al. [37] evaluated different locations from the Flanders region of Belgium against uncertain energy markets and proposed a location that shows better indicators against uncertainties.

1.3. Novel contributions

In most previous studies on techno-economic analysis and optimization of 5GDHC networks, uncertain variations in energy market scenarios and demands have often been overlooked. Only a few studies, such as [3,32–34], have considered the analysis of network design under the influence of uncertainties. For instance, in [32], a Monte Carlo method is employed alongside commercial tools like TRANSYS and Fluidit-heat to assess the techno-economic performance of 5GDHC networks amidst uncertainties. Similarly, [33] conducted a risk assessment concerning economic indicators for the network and compared it with individual systems. However, neither study performed robust design optimization for sustainable operations under uncertain future scenarios. The study conducted in [3] is quite similar to the proposed work by providing five step methodology for uncertainty and sensitivity analysis of the DHC systems. For each uncertain scenario, an optimization routine based on mixed integer linear programming is used to determine the optimal network design and operational parameters. Subsequently, a probability distribution is computed for performance indices such as the levelized cost across different uncertainties. However, their approach does not differentiate which individual input exerts a greater impact on the performance index; instead, it provides a probability density function (PDF) considering the combined effect of all uncertain inputs. Another limitation of the method is the utilization of linear models, which may not ac-

curately capture the nonlinear behavior arising from thermo-hydraulic changes inherent in 5GDHC networks, as discussed in [5].

Abokersh et al. [34] offers a comprehensive Global Sensitivity Analysis (GSA) utilizing Sobol's indices for a solar-assisted district heating (DH) network. Furthermore, they devised an Artificial Neural Network (ANN)-based meta-model to depict the entire system, along with the variance-based model. However, their approach overlooks the consideration of bi-directional flows, which have a significant impact, especially in the case of 5GDHC networks operational design and optimization. This work addresses current research gaps by applying machine learning to computationally expensive high-fidelity models of network simulations for heat and hydraulics. A preliminary sensitivity analysis is conducted to identify design variables and objectives, facilitating feasible and computationally efficient solutions for multi-objective problems, as demonstrated in [3,38,39]. Thus, this study not only addresses gaps in previous approaches but also provides a systematic method to identify potential 5GDHC cases. The generic thermodynamic framework from the authors' previous work [37] is extended to align with a nonlinear model-based robust design approach to enhance the potential applicability of business cases by improving the components of financial metrics.

The novel contributions of this paper are summarized as:

- Generation of uncertain energy market and demand scenarios through probabilistic descriptions
- Framework to assess thermodynamic and economic performance under uncertainty
- Identification of promising business cases through risk analysis of financial metrics for business case
- Machine learning based on KPI responses obtained from high-fidelity models
- Solving multi-objective robust design optimization (MO-RDO) problem to find Pareto-optimum solutions for network designs
- Evaluating the risk on financial metrics with the solved MO-RDO

1.4. Paper structure

The rest of the article is organized as follows: in section 2, we first explain the general scope of the work and the importance of robust design analysis and optimization under the influence of uncertain markets, which hinder the transition towards 5GDHC networks. In section 3, a general thermodynamic framework for evaluating potential 5GDHC network business cases, following their preliminary assessment using GIS-based methods. The section also explains how network KPIs are defined to compute financial metrics in comparison to present-day design technologies. Additionally, it addresses uncertainties from energy markets and their impact on financial metrics, along with methods for risk evaluation. Furthermore, we develop machine learning algorithms to create a surrogate model that can initially identify the influential design variables and objectives prior to optimization. This model is then utilized in the robust optimization problem to find optimal trade-offs between environmental and economic objectives. In section 4, first, a small virtual network is considered to present year-based KPIs correlation over financial metrics, followed by a small business case of 5GDHC is analyzed using the proposed methodology. In section 5, the simulation results are displayed, indicating which input uncertainties should be considered from a network operational and design perspective, and the optimal trade-offs determined. We discuss the outcomes and their applicability to provide insights for investors. Section 6 presents the conclusions of this study by highlighting the achievements yielded by the proposed methodology.

2. Problem statement

The transition to 5GDHC networks faces limitations due to uncertainties in present-day energy markets, constraining investors from making

decisions even when a particular site appears attractive. These uncertainties encompass factors such as equipment costs, energy prices, and CO₂ taxes. Moreover, in many 5GDHC networks, energy is exchanged among network participants, amplifying the impact of uncertainties in participant demands and further restricting the transition. Importantly, a preliminary analysis using GIS-based approaches, as discussed in [27,40–42], is insufficient to address the uncertainty effects related to unpredictable energy market behavior and fluctuating energy demands.

In this work, a systematic methodology is developed, beginning with an analysis of potential locations using simple linear and randomly generated scenarios to assess the techno-economic performance over the project's life, focusing on financial metrics to assess the profitability and return on investment. Locations that demonstrate stable KPIs are then utilized for robust design optimization, aiming to identify environmentally or economically efficient solutions. These optimized solutions are intended to improve overall financial metrics while accounting for uncertainties that may hinder the transition process. This will assist investors in making informed decisions based on financial analysis.

3. Methodology

This section presents the methods and framework to evaluate and optimize the business case for 5GDHC networks in business parks in Flanders, Belgium. It considers uncertain energy market and demand scenarios. The framework, shown in Fig. 1, consists of four steps.

Step 1 explains the thermodynamic modeling framework section 3.1, used to analyze locations identified through a GIS-based approach. It also details how prosumers' energy demands are modeled and how network equipment is selected. The operational costs of the equipment are also discussed. In section 3.2 and section 3.3, the characterization of KPI metrics and uncertain input scenarios are explained, respectively.

Step 2 involves risk analysis discussed in section 3.3.1, assessing the project's net present value against uncertain energy market scenarios. Based on Key Performance Indicator (KPI) responses, a preliminary techno-economic assessment is conducted to rate each location's future effectiveness in energy demand sharing.

In Step 3, the machine learning approach detailed in section 3.4 is applied to analyze over 3000 uncertain input scenarios, identifying the key factors that influence the KPIs of the 5GDHC network. This simplifies the multi-objective optimization process in Step 4, discussed in section 3.5, where trade-offs between environmentally and economically optimal solutions are examined using a computationally efficient model based on the PC-Kriging approach.

3.1. 5GDHC network modeling

After conducting the preliminary analysis using a GIS-based approach in Step 1, the most promising locations are shortlisted. In a GIS-based approach, various types of data are uploaded onto platforms like QGIS to evaluate the potential of specific locations. This data may include historical information about how people or systems have interacted, current network layouts of utilities like gas and electricity, basic information about the landscape, and weather data. The shortlisted sites are then further analyzed in Step 2 using the Python-based Heat Distribution System Simulator (HDSS) for detailed design.

3.1.1. Assumption and approach for 5GDHC analysis

The HDSS modeling layout for representing the 5GDHC network description is illustrated in Step 2 on the right side of Fig. 2. The HDSS considers the following assumptions to perform network simulations:

- Thermal losses through the main network pipes are negligible due to the proximity of network temperatures to ambient levels.
- The hydraulic calculations are performed by assuming a fixed friction factor for pipe hydraulic losses.

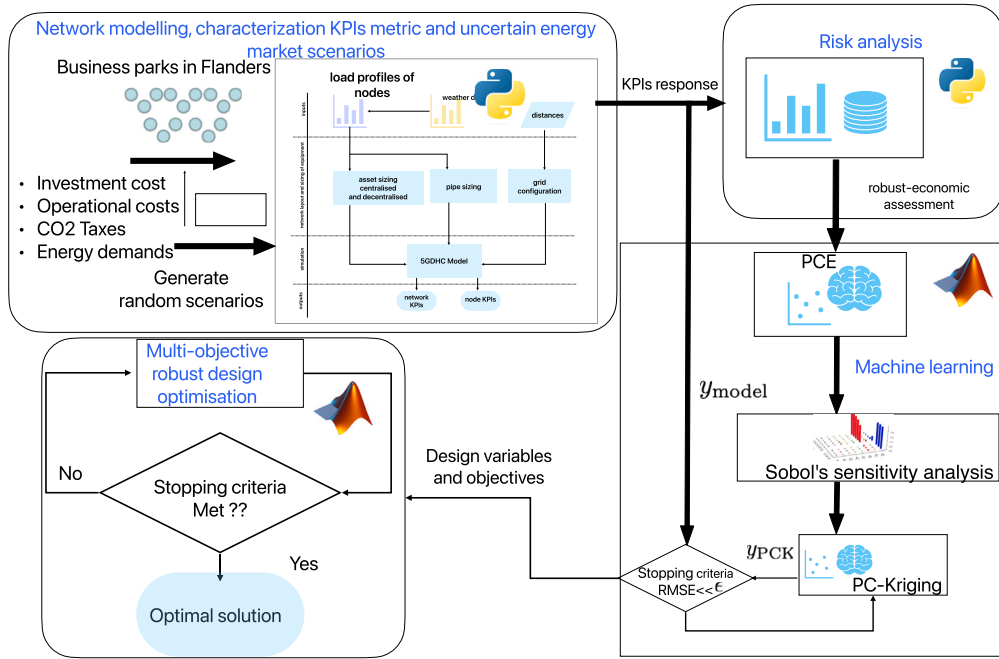


Fig. 1. A framework for 5GDHC networks robust design optimization.

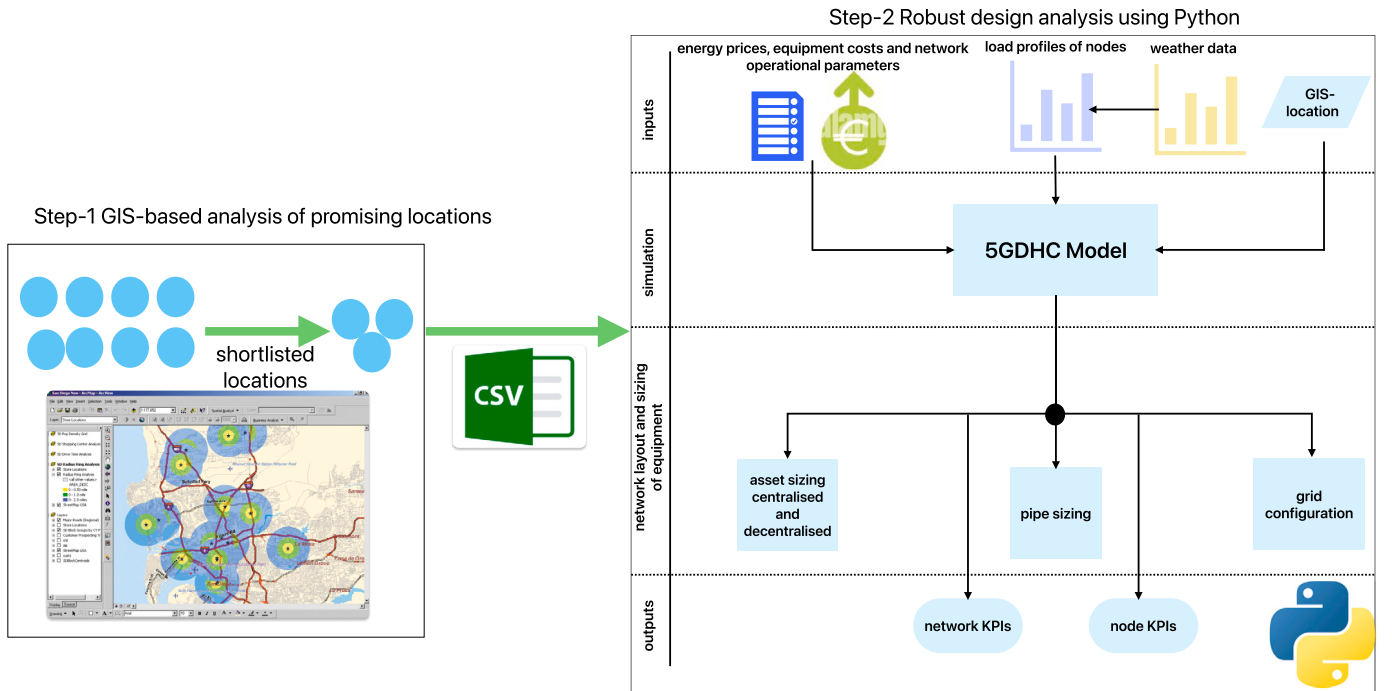


Fig. 2. Schematic layout for assessing potential 5GDHC locations using a combined GIS and Python-based approach.

- The cost of network connection and maintenance are not considered.
- The temperature of hot and cold pipes is fixed for 1 y of simulation.
- The efficiency of natural gas (NG) boiler (η_{boiler}) is fixed to 90 %.

The HDSS exchanges information with the GIS database using a CSV exchange format, listed in Table 1. In this format, table columns x and y represent the position of a prosumer node in meters within the Earth's coordinate system; column SA represents the building surface area, and column Q represent the energy demand over the year. Another exchange format given in Table 2, provides details about the connection between

Table 1

Data exchange format from the GIS to the HDSS, listing the position, surface area, consumption, and waste heat production.

ID	Name	Q [MWh]	SA [m ²]	x [m]	y [m]
N01	data center	xxxx	xxxx	xxxx	xxxx
N02	office	xxxx	xxxx	xxxx	xxxx
....		xxxx	xxxx	xxxx	xxxx

the prosumers. Each participating prosumer node, $n \in (1, N)$ has a connection with the network via a network pipe, $D_{i,j}$ with $ij \in (1, N)$,

Table 2

Data exchange format from the GIS to the HDSS, listing the size of each network section.

ID	From	To	D [mm]
D01	data center	office1	xxx
D02	office1	office2	xxx
....			xxx

Table 3

List of parameters used by HDSS to simulate 5GDHC cases, these parameters are used to simulate different input scenarios.

descriptions	symbol	value	unit
energy demand	$\dot{Q}_{h,c}$	–	kW
NG boiler efficiency	η_{boiler}	90	%
BHP efficiency	η_{BHP}	70	%
GSHP efficiency	η_{GSHP}	50	%
ASHP efficiency	η_{ASHP}	40	%
hot network temperature	T_h	35	°C
cold network temperature	T_c	20	°C
outdoor temperature	T_{out}	–	°C
indoor temperature	T_{in}	–	°C
ASHP factor	h_1	9817	–
GSHP factor	h_2	25051	–
BHP factor	h_3	25051	–
BTES fix	c_{BTES}	16726	EUR/m ³
ATES fix	c_{ATES}	145	EUR/m ³
boiler price	c_{boiler}	100	EUR/kW
trench price	c_{trench}	1.0	kEUR/m
gas price	c_{gas}	0.05	EUR/kWh
electricity price	c_{el}	0.10	EUR/kWh
CO ₂ penalty	c_{CO_2}	0.15	EUR/kgCO ₂
CO ₂ emission electric	e_{el}	0.24	kgCO ₂ /kWh
CO ₂ emission gas	e_{gas}	0.21	kgCO ₂ /kWh
slope hot line	S_h	1.0	°C/°C
slope cold line	S_c	0.2	°C/°C

where i and j indicates the table columns ‘From’ and ‘To’ node. It then uses this information to draw and size the network and to analyze energy flows within the network by employing graph theory [15].

It is important to note that synthetic load profiles for participating nodes, based on weather variations, are prepared priorly using the methodology described in section 3.1.2. The HDSS collects load profile data from participating buildings or prosumers (referred to as external nodes of the network), including their GIS locations, energy costs, investment costs, and network temperatures, to simulate the network as shown in Fig. 2. It then uses the operational data and investment costs listed in Table 3 to simulate and compute the key performance indicators (KPIs) for both the prosumer and network operators.

3.1.2. Building demands modeling

In this section, the methods for estimating the energy demands of participants in the 5GDHC network across different seasons are discussed. The methodology from the authors’ previous work [43] was applied to estimate prosumer energy demands, accounting for seasonal variations. Outdoor temperature data was sourced from the Royal Meteorological Institute in Belgium.

Hachez et al. [43] uses the energy data related to electric and gas consumption (non-HVAC) of the prosumer to train the building energy model. This dataset includes the last five years’ consumption from 2018 to 2022 for prosumer-type buildings such as indoor skiing facilities, supermarkets, and data centres. The methodology requires building details such as type of building, annual consumption Q_{building} , buildings’ energy performance certificates (EPC), surface area (SA), consumption factors for weekends and weekdays over a year and outdoor temperature profile to separates HVAC energy data related to outdoor weather conditions from the non-HVAC-related energy data using stochastic approach as

discussed in [43]. The annual consumption for a given EPC level and SA can be computed using

$$Q_{\text{buildings}} = \text{EPC} \cdot \text{SA}. \quad (1)$$

Energy profiles are generated at an hourly resolution for the period spanning from YYYY:MM:DD hr:min:sec to YYYY:MM:DD hr:min:sec, corresponding to the total annual consumption ($Q_{\text{buildings}}$). The detailed temporal breakdown provides insights into the variation of energy demand across the entire year as discussed in [44–46]. Interested readers are referred to [43] for more information on prosumer energy demand profile generation.

3.1.3. Sizing of network equipment and associated costs

It is important to mention that in HDSS, each node is referred to as a decentralized prosumer node and has been sized according to the peak loads. The decentralized prosumer node or substation consists of a BHP that provides hydraulic isolation between primary and secondary hydraulic circuits [47]. In decentralized prosumer node operations, a BHP is utilized, connected to both the hot and cold network, as depicted in Fig. 3. Additionally, the centralized assets, referred to as the balancing unit and network pipes in HDSS, are sized based on the variance in energy demands among participating nodes as shown in Fig. 3 on the right. This concept is also applied in [7], where supplementary generation units are added when borehole energy is insufficient to meet all demands.

For the sake of simplicity, we assumed that the balancing unit is equipped with a chiller, a natural gas (NG) boiler, and a lossless thermal energy storage. In the context of managing balancing units and network operations, we assume that the assets work at low temperatures and that the maximum part of the network imbalance should be covered by seasonal storage. In this study, two kinds of seasonal storage are considered; one is called borehole thermal energy storage (BTES) and the other is called aquifer (ATES) [48]. The volume of the storage can be calculated as [49]:

$$V_{\text{storage}} = \frac{Q}{\rho \cdot c_p \cdot \Delta T}, \quad (2)$$

where ΔT is the temperature difference of working fluid in the supply network (T_h) and cold network (T_c). As observed from eq. (2), temperatures in hot and cold pipes close to ambient will require a larger TES volume to cover the mismatch in heating and cooling demand over a season.

The specific costs for two types of storage are given in [49] as:

$$C_{\text{specific,borehole}} = c_{\text{BTES}} \cdot V_{\text{storage}}^{-0.7}, \quad (3)$$

$$C_{\text{specific,aquifer}} = c_{\text{ATES}} \cdot V_{\text{storage}}^{-0.1}. \quad (4)$$

The KPIs related to investment cost are then calculated using

$$\text{capex}_{\text{storage borehole}} = C_{\text{specific,borehole}} \cdot V_{\text{storage}}, \quad (5)$$

$$\text{capex}_{\text{storage aquifer}} = C_{\text{specific,aquifer}} \cdot V_{\text{storage}}. \quad (6)$$

The cost of network pipe [EUR/m] varies with diameters D_i and trench cost as discussed in [3]. For the network connections, the considered cost includes the investment cost for the pipe, which also encompasses the excavation or trench cost. This corresponds to the expenses associated with digging and preparing a trench for various purposes, such as laying pipes, cables, or other utilities and can be calculated as:

$$\text{capex}_{\text{pipe},i} = 2.16 \cdot D_i + \text{trench price} + 68. \quad (7)$$

Those investment costs are summed on every pipe and constitute the investment cost of pipes at the network level as CAPEX_c, and the associated cost of the pumping power to maintain energy flows through the pipes is summed as OPEX_c, as shown in Fig. 3.

It is important to note that, for comparing savings or revenue with the network design scenario, current technologies like NG boiler

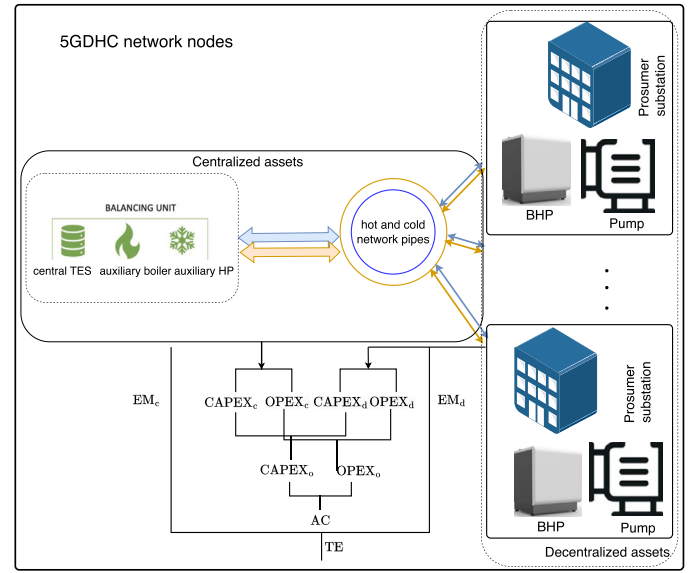
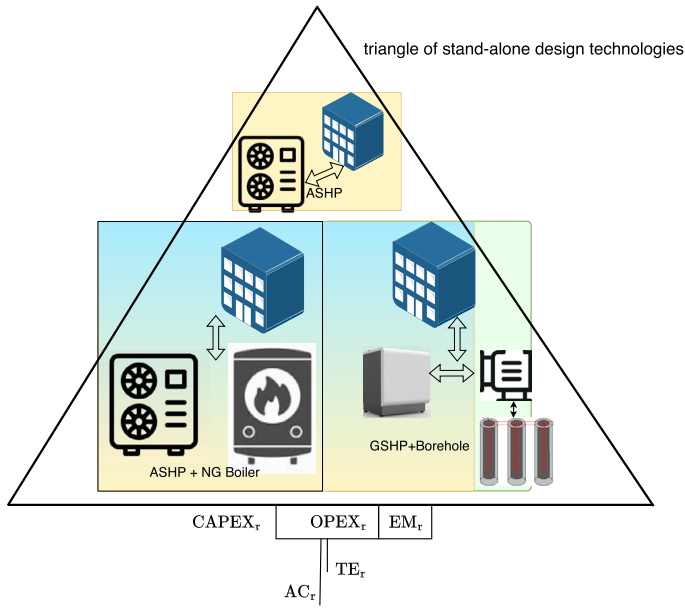


Fig. 3. Cost addition scheme for 5GDHC network node scenario compared to standalone design technologies: on the left, a triangle representing standalone technologies; on the right, cost evolution from individual assets to total 5GDHC network costs.

ers, GSHPs, and ASHPs are considered, as discussed in [32]. The sizes and associated investment costs of the standalone design technologies, depicted on the left side of Fig. 3, for meeting heating and cooling demands using ASHPs, GSHPs, and boilers, can be calculated as:

$$\text{CAPEX}_{\text{tec}} = c_t \cdot \max(\dot{Q}) + h_{\text{tec}} \quad \text{tec} \in [\text{BHP}, \text{ASHP}, \text{GSHP}, \text{NG}], \quad (8)$$

where $h_{\text{tec}} = [25051, 9817, 25051, 0]$ is the vector of unitless regression factors for estimating capacity factors of considered technologies.

In the case of a GSHP, a small-scale borehole comprised of a horizontal tube-based ground heat exchanger to satisfy its heating and cooling demands over the seasons is also considered ([50,51]). In [51], a method is constructed to estimate the ground heat exchanger sizing and thermal properties. The desired lengths of the ground-based heat exchangers concerning the heating and cooling demands of the nodes are given as discussed in [51]

$$L_h = \frac{\dot{Q}_{h,\max}(K_p + K_s F_h)}{T_{\text{out}} - T_c} \cdot \frac{\text{SCOP}_h - 1}{\text{SCOP}_h}, \quad (9)$$

$$L_c = \frac{\dot{Q}_{c,\max}(K_p + K_s F_c)}{T_h - T_{\text{out}}} \cdot \frac{\text{SCOP}_c - 1}{\text{SCOP}_c}, \quad (10)$$

where K_p and K_s are the pipe and soil resistances, F_h and F_c are the run fractions, and given in [51]. $\dot{Q}_{h,\max}$ and $\dot{Q}_{c,\max}$ are the peak heating and cooling demands, and SCOP_h and SCOP_c are seasonal COP of heat pumps of the form eq. (21), respectively. The required volume is calculated with the length as $V_{c,h} = \pi \cdot \frac{D_{\text{bore}}^2}{4} \cdot L_{c,h}$.

The investment cost for a ground-based heat exchanger or a borehole is calculated as

$$\begin{aligned} \text{CAPEX}_{\text{GHX}} &= c_{\text{drill}} \cdot \max(L_c, L_h) + \\ & c_{\text{tube}} \cdot \max(L_c, L_h) + c_{\text{tube}} \cdot V_{c,h}. \end{aligned} \quad (11)$$

The overall investment cost of GSHP ($\text{CAPEX}_{\text{GSHP},0}$) case can be calculated using

$$\text{CAPEX}_{\text{GSHP},0} = \text{CAPEX}_{\text{GSHP}} + \text{CAPEX}_{\text{GHX}}. \quad (12)$$

3.1.4. Operations of 5GDHC substation against modern design technologies

The COP of each heat pump technology can be determined by adjusting the theoretical maximum $\text{COP}_{\text{Carnot}}$ with the heat pump exergy efficiency (η_{HP}) [50,52] for $i \in [\text{BHP}, \text{ASHP}, \text{GSHP}]$

$$\text{COP}_{\text{HP},i} = \eta_{\text{HP},i} \cdot \text{COP}_{\text{Carnot}} \quad i \in [\text{BHP}, \text{ASHP}, \text{GSHP}]. \quad (13)$$

The $\text{COP}_{\text{Carnot}}$ of the heat pump can be calculated as [53]:

$$\text{COP}_{\text{Carnot}} = \frac{T_{\text{cond}}}{T_{\text{cond}} - T_{\text{evap}}}, \quad (14)$$

where the condenser operates at $T_{\text{pinch}} = 5^\circ\text{C}$ above the emitter temperature (T_{emit}). The evaporator operates at $T_{\text{pinch}} = 5^\circ\text{C}$ below the lowest temperature (T_{low}) in grid [54]. This results in eq. (14) as follows

$$\text{COP}_{\text{Carnot}} = \frac{T_{\text{emit}} + T_{\text{pinch}}}{T_{\text{emit}} - T_{\text{low}} + 2T_{\text{pinch}}}, \quad (15)$$

and the T_{pinch} is the static temperature difference between refrigerant and working fluid [39]. The emitter temperature (T_{emit}) of heat pump can be calculated as:

$$T_{\text{emit}} = T_{\text{in}} + S(T_{\text{in}} - T_{\text{out}}). \quad (16)$$

The emitter temperature depends on the outside temperature and the slope S , where the value of S ranges from $0.75^\circ\text{C}/^\circ\text{C}$ to $1.75^\circ\text{C}/^\circ\text{C}$ to supply the heat to the building heating system at varying temperatures [55]. Similarly, the equivalence to compensate for the heat gains, the cold side temperature (T_{low}) is given as

$$T_{\text{low}} = T_{\text{in}} + S_c(T_{\text{out}} - T_{\text{in}}). \quad (17)$$

$S_c = 0.2$ represents the situation when the T_{in} is fixed at 6°C , and building needs cooling.

Due to significant temperature variations at the inlet of an air source heat pump (ASHP) compared to a fixed ground water source heat pump (GSHP), the η_{HP} is lower for ASHPs than for GSHPs, as discussed in [52]. In Table 4, the characterization of T_{emit} and T_{low} temperatures and η_{HP} are listed with respect to different design technologies and in different modes of operations. The eq. (13) can be modified with these temperatures across the HPs and their η_{HP} to compare the techno-economic performance between different design technologies.

Table 4

List of temperatures to compute COP of HPs for competing design technologies scenarios (BHP, ASHP, GSHP).

mode	scenario	η_{HP} [%]	T_{emit} [°C]	T_{evap} [°C]
heating	BHP	70	$T_{h,dem}$	$T_{c,grid}$
	GSHP	50	$T_{h,dem}$	$T_{out,ground-water}$
	ASHP	40	$T_{h,dem}$	$T_{out,air}$
cooling	BHP	70	$T_{h,grid}$	$T_{c,dem}$
	GSHP	50	$T_{out,ground-water}$	$T_{c,dem}$
	ASHP	40	$T_{out,air}$	$T_{c,dem}$

The time series for electric consumption of the heat pump ($\dot{W}_{HP,i}$) in heating and cooling modes can be computed using the standard form of COP_{HP,i} eq. (13) as:

$$\dot{W}_{HP,i,h} = \frac{\dot{Q}_h}{COP_{HP,i}}, \quad (18)$$

$$\dot{W}_{HP,i,c} = \frac{\dot{Q}_c}{COP_{HP,i} - 1}. \quad (19)$$

Whereas, depending upon the mode of operations and considered design technology, the COP_{HP,i} is computed with the relevant temperatures and efficiencies as listed in Table 4. The annual consumption of the pump is then calculated as:

$$W_{HP,i} = \sum_{N=1}^{N=8760} \dot{W}_{HP,i,hc} \quad (20)$$

The seasonal COP of heating and cooling can be computed as:

$$SCOP_{hc} = \frac{\sum_{N=1}^{8760} (\dot{Q}_{hc})}{\sum_{N=1}^{8760} (\dot{W}_{hc})}. \quad (21)$$

OPEX for the HP technologies ($i \in [\text{ASHP, GSHP, BHP}]$) are calculated as:

$$OPEX_{HP,i} = c_{el} \cdot W_{HP,i}, \quad (22)$$

and the emissions related to operations are calculated as

$$EM_{HP,i} = c_{el} \cdot W_{HP,i,hc}, \quad (23)$$

$$C_{em,HP,i} = c_{CO_2} \cdot EM_{HP,i}. \quad (24)$$

The gas consumption of the boiler is computed as

$$Q_{gas} = \sum_{N=1}^{N=8760} \dot{Q}_{gas}, \quad (25)$$

whereas hourly gas consumption of the boiler can be computed with $\eta_{boiler} = 90\%$ as

$$\dot{Q}_{gas} = \frac{\dot{Q}_{build}}{\eta_{boiler}}. \quad (26)$$

The economic KPIs (opex/y and capex/y) for boiler case are calculated as:

$$CAPEX_{boiler} = c_{boiler} \cdot \max(\dot{Q}_{gas}), \quad (27)$$

$$OPEX_{boiler} = c_{boiler} \cdot Q_{gas} \quad (28)$$

The emissions-related KPIs are calculated as

$$EM_{boiler} = c_{gas} \cdot Q_{gas}, \quad (29)$$

$$C_{em,boiler} = c_{CO_2} \cdot EM_{boiler}. \quad (30)$$

Table 5

List of main KPIs metrics.

no.	description	symbol	units	type
1	synergy	–	%	technical
2	5GDHCness	RV	%	technical
3	capex overall	CAPEX _o	kEUR/y	economic
4	net savings	χ	kEUR/y	economic
5	cost ratio	CR	%	economic
6	savings-to-capex	SC	%	economic
7	total emissions	TE	tonCO ₂ eq/y	environmental

3.2. Characterization of analysis metrics

Three KPI metrics are used to analyze the potential 5GDHC network design scenario, covering technical, economic and environmental aspects. HDSS computes these KPIs at a hierarchical level on a yearly basis, for instance, KPIs related to node level and network level using the methods discussed in section 3.1. These KPIs are accumulated throughout the simulation over the project's lifespan. The HDSS aggregates these yearly evaluations to assess the financial indicators over the project's lifespan, as given in Fig. 4.

3.2.1. Yearly based KPIs assessment

The main technical, economic and environmental KPIs that are used in this study are listed in Table 5. These KPIs are computed based on the methods explained in section 3.1. For instance, synergy is the technical KPI showing the overlapping amongst the network participants [26]. In a GIS-based tool, synergy is a good predictor for finding the potential of 5GDHC [27]. Based on the stochastic profile estimation methodology discussed in section 3.1.2, we compute the synergy between participants of the network. Synergy 0% means no matching, and 100% means that energy demands are fully overlapped over the entire seasons.

Reversibility or reverse flow (RV [%]) is the percentage of the time flow moves in the opposite direction of the dominant flow and can be computed as:

$$RV = \left(\frac{t_{reverse}}{t_{total}} \right) \cdot 100, \quad (31)$$

where $t_{reverse}$ represents the amount of time that energy flows in the opposite direction due to cooling demand compared to the total flow duration t_{total} .

In section 3.1.3, based on the energy demands profile of the prosumer, HDSS selects the size of the equipment and calculates the investment costs of centralized and decentralized equipment (CAPEX_c and CAPEX_d) using eqs. (5) to (8). Operational cost and associated CO₂ tax of assets are calculated in section 3.1.4 using eqs. (22) and (24). It can be seen from Fig. 3 that all the network upfront investment costs and operational expenditures, including CO₂ tax, are summed using eqs. (32) and (33), respectively.

$$CAPEX_o = CAPEX_c + CAPEX_d, \quad (32)$$

$$OPEX_o = OPEX_c + OPEX_d. \quad (33)$$

This leads to annualized overall cost (AC) as

$$AC = CAPEX_o + OPEX_o, \quad (34)$$

Similarly, the emissions associated with the centralised and decentralised equipment and CO₂ tax can be calculated as

$$TE = EM_c + EM_d, \quad (35)$$

where HDSS calculates centralized emissions EM_c and decentralized emissions EM_d using eqs. (23) and (29).

As discussed above, some of the KPIs indicating the economic performance of the 5GDHC networks are related to the KPIs of present-day design technologies such as boilers, air source heat pumps (ASHP), and

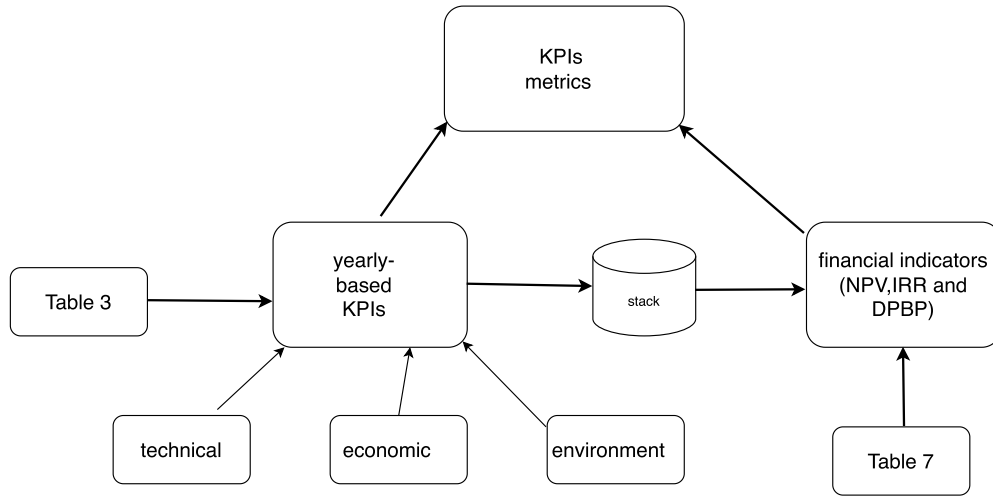


Fig. 4. Characterization of KPI-metrics to evaluate potential business case.

ground source heat pumps (GSHP). This study compares the 5GDHC network scenario against ASHP + boiler technology scenario to compare financial performance indicators. When comparing boilers and chillers with the 5GDHC network, we assume that building substations work independently and aren't connected to the network. The assumption holds for other design technology scenarios as well. The main difference would be that HDSS will select a relevant set of equations for a particular technology scenario described in sections 3.1.3 and 3.1.4 to compute $CAPEX_r$ and $OPEX_r$. The overall annualized cost with reference technology is then calculated as

$$AC_r = CAPEX_r + OPEX_r, \quad (36)$$

and TE_r can be calculated by summing all emissions associated with reference design technologies ($\sum EM_r$). Whereas HDSS calculate EM_r using eqs. (23) and (29), respectively.

The net operational cost savings (χ) calculated in eq. (37), derived from network operations, is compared with the operational expenditures ($OPEX_r$) eq. (28) associated with a boiler's reference design for heating and considering ASHP operational cost for cooling using eq. (22).

$$\chi = OPEX_r - OPEX_o, \quad (37)$$

$$CR = \frac{AC}{AC_r}, \quad (38)$$

$$SPBP = \frac{CAPEX_o - CAPEX_r}{\chi}. \quad (39)$$

When $\sum(OPEX_o) < OPEX_r$, it indicates positive net savings ($\chi > 0$ EUR/y). Both $CAPEX_o$ and χ directly influence the component of financial metrics, such as net present value (NPV), as discussed in [47, 18]. The cost ratio (CR) is represented by the ratio of AC to AC_r . A $CR \leq 100\%$ is considered a favourable indicator for a 5GDHC network. In addition to CR, another economic-KPI savings-to-capex ($\frac{\chi}{CAPEX_o}$) ratio is considered. A higher savings-to-capex ratio enhances financial KPIs and improves the project's feasibility over its lifetime. The simple payback period (SPBP), as given in eq. (39), depends on the ratio between the CAPEX and χ . Since SPBP and χ are inversely related, a higher χ leads to a shorter SPBP for a given CAPEX.

3.2.2. Financial KPIs

Reflecting on the primary objective of the proposed framework, it aims to optimize financial KPIs for environmentally or economically efficient solutions. Key indicators like net present value (NPV), internal rate of return (IRR), and discounted payback period (DPBP) help investors assess a site's feasibility by comparing initial investment with cash flows and operational cost savings. NPV serves as a financial KPI for locations,

allowing the estimation of costs and benefits in cash-equivalent terms. Interesting business cases require NPV higher values to enhance project desirability. The NPV for the 5GDHC case is calculated as ([56,57]):

$$NPV = \sum_{n=0}^N \frac{CF_n}{(1+r)^n}, \quad (40)$$

where r represents the discounted rate on investments, CF_n represents net cash flows over N years, $CF_0 = CAPEX_o$ and $CF_{1,...,n}$ correspond to $\chi_1, \dots, \chi_n$ against certain increase in market inflation ([47]). In this study, market inflation is considered for the base case scenario reported in PowerPulse, see for example Table 6. The Table 6 indicates that electricity and gas prices vary $\approx 1\%$ per year (as inflation) over the project life of 40 y.

Internal rate of return (IRR [%]) measures the investment profitability and higher IRR implies more desirable investments ([56]). IRR is the value of discount rate which gives $NPV = 0$ as :

$$\sum_{n=0}^N \frac{CF_n}{(1+IRR)^n} = 0. \quad (41)$$

DPBP measures the discounted payback period in years for different locations. In simple terms, it indicates the period at which $NPV > 0$, as:

$$DPBP = \min \{ N \mid \sum_{n=0}^N \frac{CF_n}{(1+r_{nom})^n} \geq 0 \}, \quad (42)$$

DPBP exhibits greater robustness compared to a simple payback period, as it includes a discount rate on cash flows. Calise et al. [24] highlighted the long DPBP for 5GDHC networks. Therefore, in this study, a DPBP shorter than the equipment replacement time, such as under 20 y, is considered a more suitable indicator.

To evaluate all the aforementioned financial metric components, the operation and maintenance (O&M) cost is treated as a fixed replacement cost (c_{rep}) on the initial investment ([3]). Some corporation tax (c_{tax}) has also been included in the net cash flows. The c_{rep} will apply after some years of the project; Saini et al. [32] considered 15 y. However, in this study, we apply this cost after half of the project's lifespan. Moreover, c_{tax} is applied from the beginning of the project and will continue until the end of the project's life. The parameters used to calculate lifecycle cost economics are listed in Table 7.

3.3. Characterizing uncertainties and scenarios for risk analysis

The uncertainties in energy prices, CO₂ tax, equipment costs, and energy demands of the buildings are three major types of uncertain-

Table 6
Electricity and gas price variations for three different categories, opted from PowerPulse.

price track	year	electricity price (c_{el}) [EUR/MWh]	gas price (c_{gas}) [EUR/MWh]
high gas/CO ₂ prices	2025	223	78
	2030	155	61
	2035	128	72
	2040	128	78
	2045	128	87
	2050	128	96
base	2025	144	51
	2030	105	45
	2035	91	53
	2040	91	60
	2045	91	66
	2050	91	73
low gas/CO ₂ prices	2025	93	36
	2030	70	28
	2035	64	39
	2040	64	47
	2045	64	49
	2050	64	52

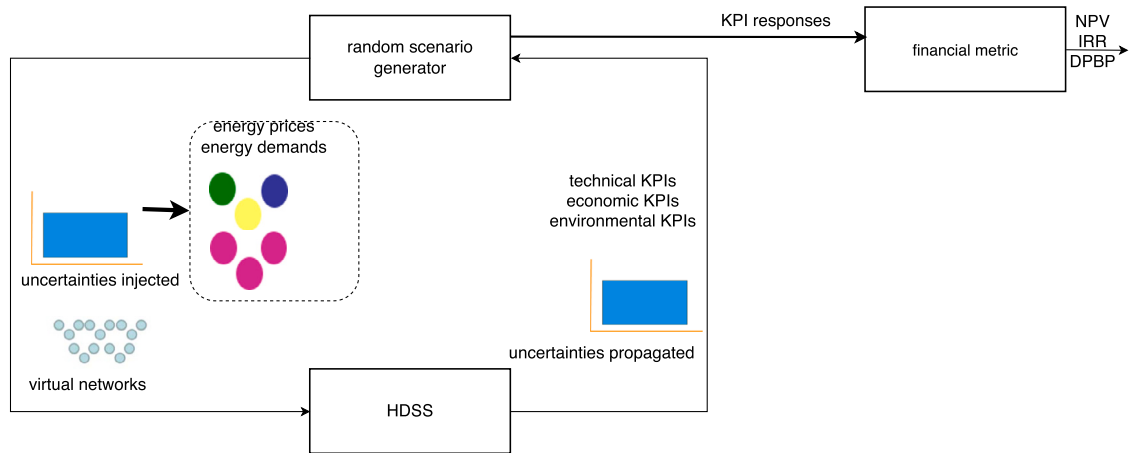


Fig. 5. Proposed scheme to evaluate the 5GDHC network's KPI metrics.

Table 7
List of parameters used to analyze the financial metrics.

descriptions	symbol	value	unit	source
discount rate	r	2 to 12	%	–
life span	N	40	y	as per pipe and TES
start year	N_{start}	2024	y	–
end year	N_{end}	2063	y	–
first replacement	k	20	y	as per heat pumps
corporation tax	c_{tax}	1	%	assumed, start $N = 1$ y
replacement cost	c_{rep}	4	%	assumed, start $k = 21$ y

ties originating from the energy market ([3]). To represent the characteristics of future changes in equipment cost and energy prices, the probabilistic density function, which explains the range of values and the probability of occurrence, is commonly adopted ([58]). However, if there is no explicit reason to value one probability distribution over another, a uniform distribution can be used as discussed in ([3]). Hence, the uniform distribution is applied to input parameters related to equipment costs and energy prices as given in Table 3. The uniform distribution of these parameters is listed in Table 8. The random scenario generator block shown in Fig. 5 injects uncertainty in the inputs and stacks the propagated uncertainty as KPIs response.

The energy demand variations in buildings are influenced by changes in thermodynamic properties that can occur over the long project life [3, 16]. Consequently, the buildings' energy performance certificates (EPC)

can change over time. In this study, these variations are considered to be uniformly distributed with EPC = 0.1 to 0.4 MWh/m²y, representing buildings with performance labels ranging from A⁺ to E for a given surface area (SA). The random scenario generator injects uncertain EPC levels to update the annual consumption of the building using eq. (1) and Table 1 before HDSS step execution.

The network's supply and return temperatures, as well as the heat pump's supplying temperature to prosumer premises, may vary due to uncertainties in the entire energy demands [8,43]. Therefore, the network's operational parameters and pipe sizes, detailed in Table 9, are also regarded as uncertain inputs to assess their impact on KPIs.

In this study, uncertainties in the parameters used to calculate financial metrics over the project lifespan, as listed in Table 7, were not considered. For instance, the discount rate is not uniformly distributed as higher discount rates diminish the value of the initial investment. Adding more parameters would also reduce the framework's computational efficiency.

3.3.1. Risk analysis

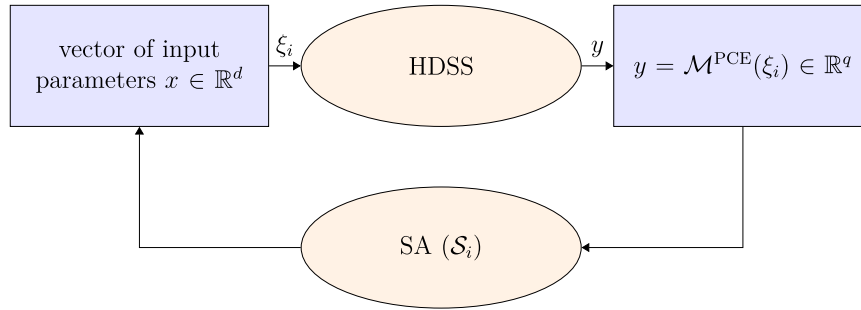
The risks are evaluated with statistical parameters such as data histogram, probability density function, cumulative density function and quantiles of NPV. To calculate PDF an exponential distribution is selected as:

$$f(x; \lambda) = \lambda e^{-\lambda x}, \lambda = f(\mu, \sigma), \quad (43)$$

Table 8

List of input parameters representing bounded uncertain energy market variations.

input parameter	description	initial value	final value	units
electricity price	price variation, PowerPulse	0.15	0.09	EUR/kWh
gas price	price variation, PowerPulse	0.05	0.07	EUR/kWh
emission electricity	carbon intensity from Electricity MAPS	0.02	0.3	kgCO ₂ /kWh
CO ₂ penalty	GHG emission tax	0.10	0.40	EUR/kgCO ₂
ASHP price	assumed capacity price for individual building equipment	145	155	EUR/kW
GSHP price	assumed capacity price for individual building and network substation	103	113	EUR/kW
BTES price	capacity price for seasonal storage as per FlexyNet	16	18	kEUR
ATES price	capacity price for seasonal storage as per FlexyNet	145	155	EUR
boiler price	assumed capacity price for individual building equipment	100	200	EUR/kW
trench price	assumed capacity price for network pipes	1	2	kEUR/m

**Fig. 6.** Schematic diagram to perform global sensitivity analysis using variance-based method.**Table 9**

Uncertainties in the 5GDHC network design and operational parameters.

description	symbol	min	max	unit
pipe diameter	d_{ij}	20	1400	mm
hot temperature	T_h	11	35	°C
cold temperature	T_c	1	10	°C
slope cold line	S_c	0.1	0.3	°C/°C
slope hot line	S_h	0.8	1.7	°C/°C

where the CDF is calculated as :

$$F(x) = \int_{-\infty}^x f(t) dt, \lim_{x \rightarrow \infty} F(x) = 1. \quad (44)$$

Using Fig. 5, the KPI responses for a potential location are generated over the project life against the uncertain combination of inputs, and then NPV is calculated in post-processing.

3.4. Machine learning with KPI responses

In this section, machine learning with the response of a Python-based heat distribution system simulator (HDSS) is developed. The developed ML model is then used in the subsequent steps, such as design variables and objectives of robust optimization through global sensitivity analysis (Sobols indices). Importantly the ML- model will also replace the HDSS in the formulated optimization problem.

3.4.1. Polynomial chaos expansion for global sensitivity analysis

Analyzing a complex engineering system involves understanding the functional mapping relationship between the various input variables and the considered KPIs ($q \in [\text{CAPEX}_0, \chi$ and $\text{TE}]$). We suppose the variable y has the function mapping relationship with the variable x as $y = f(x)$, where $x = (x_1, x_2, \dots, x_d)$ is a d -dimensional vector of random input variables, and $y(\xi)$ is the resulting uncertain KPI, as shown in Fig. 6. We assume $x_1, x_2, \dots, x_d$ are independent and identically uniformly distributed as listed in Table 8. The random input variables are firstly converted into standard uniformly distributed random variables

$\xi = (\xi_1, \xi_2, \dots, \xi_d)$ according to eq. (45), and the joint probability density function is $\rho(\xi)$.

$$\xi_i = \frac{x_i - \mu_{x_i}}{\sigma_{x_i}}, \quad i = 1, 2, \dots, d \quad (45)$$

where μ_{x_i} refers to the mean of the input random variable x_i of the i -th dimension, σ_{x_i} refers to the standard deviation of the input random variable x_i . Then $y(\xi)$ can be expanded to the polynomial chaos expansion as follows:

$$\mathcal{M}^{\text{PCE}}(\xi) = y(\xi) = \sum_{i=1}^{\infty} u_i \psi_i(\xi) \approx \sum_{i=1}^p u_i \psi_i(\xi), \quad (46)$$

where as p represents the polynomial order of $\mathcal{M}^{\text{PCE}}(\xi)$, and number of $\mathcal{M}^{\text{PCE}}(\xi)$ -coefficients N can be calculated as [29]:

$$N = \frac{(d+p)!}{d!p!}. \quad (47)$$

u_i denotes the PCE coefficients and $\psi_i(\xi)$ are the orthonormal basis functions concerning the probability distribution function of random variables ξ ([59]). The response can be fully determined by computing PCE coefficients (u_i) and with the help of prior knowledge of orthonormal basis functions (ψ). To compute u_i the regression method is used which selects a set of u that minimizes the mean-square error in the approximation of $\mathcal{M}$.

The coefficients vector $\hat{u}$ that minimizes the means-square error can be written as [59]

$$\hat{u} = (\psi^T \psi)^{-1} \psi^T y, \quad (48)$$

where y represents the model response vector and the ψ for a set of observations ($\xi^1, \dots, \xi^N$) can be computed as:

$$\psi = \begin{bmatrix} \psi_0(\xi^1) & \dots & \psi_i(\xi^1) & \dots & \psi_p(\xi^1) \\ \psi_0(\xi^2) & \dots & \psi_i(\xi^2) & \dots & \psi_p(\xi^2) \\ \vdots & \dots & \vdots & \dots & \vdots \\ \psi_0(\xi^N) & \dots & \psi_i(\xi^N) & \dots & \psi_p(\xi^N) \end{bmatrix}. \quad (49)$$

One of the main advantages of constructing $\mathcal{M}^{\text{PCE}}$ is to perform sensitivity analysis. The developed $\mathcal{M}^{\text{PCE}}$ is used to quantify the effect

of random vector x with d variables mentioned in section 3.3 to the corresponding output vector y by means of Sobol's as ([60]):

$$S_i = \frac{\text{Var}[\mathcal{M}_i^{\text{PCE}}(\xi_i)]}{\text{Var}(y)}, \quad i = 1, 2, \dots, d \quad (50)$$

where $\text{Var}(y)$ can be calculated as:

$$\begin{aligned} \text{Var}(y) &= \text{Var}[\mathcal{M}_i^{\text{PCE}}(x_1, x_2, x_1, \dots, x_d)], \\ &= \text{Var}[\mathcal{M}_1^{\text{PCE}}(x_1)] + \text{Var}[\mathcal{M}_2^{\text{PCE}}(x_2)] + \dots + \text{Var}[\mathcal{M}_d^{\text{PCE}}(x_d)] \quad \text{1st}^{\text{order}}, \\ &+ \text{Var}[\mathcal{M}_{12}^{\text{PCE}}(x_1, x_2)] + \dots + \text{Var}[\mathcal{M}_{d-1n}^{\text{PCE}}(x_{d-1}, x_d)] \quad \text{2nd}^{\text{order}}, \\ &+ \dots \\ &+ \text{Var}[\mathcal{M}_{12\dots d}^{\text{PCE}}(x_1, x_2, \dots, x_d)]. \end{aligned} \quad (51)$$

Hence, before performing the model-based robust design optimization only those inputs are considered which have $S_i > \epsilon$ to the selected outputs $y \in \mathbb{R}^q$.

Before implementing the optimization framework, the PCE model is expanded to incorporate a PCE-Kriging [61] modeling approach for those inputs and outputs that have $S_i > \epsilon$. As a result, a computationally efficient model can be employed by considering the design variables and objectives that are affected by input uncertainties.

3.4.2. PCE-Kriging approach

The machine-learning block is further improved to employ a PC-Kriging-based modeling approach on the modeling of influencing inputs and for selected KPIs as shown in Fig. 1. The kriging model consists of two parts such as trend and stochastic distributions [59]. The trend part reproduces the system's overall response, while the stochastic Gaussian part accounts for the residual error between the trend part and the original function. Thus, to develop an accurate meta-model of the system using the Kriging method both trend part and stochastic components need to be tuned.

In PCE-Kriging approach, the trend part is established by using PCE orthonormal as discussed in Schöbi et al. [61]

$$\mathcal{M}^{\text{PCE-Kriging}} = \beta^T f(\xi) + \sigma^2 Z(\xi, \omega), \quad (52)$$

where $\beta^T f(\xi) = \sum_{i=1}^P \beta_i z_i(\xi)$ is the trending function, $\beta = (\beta_1, \beta_2, \dots, \beta_P)$ represents the coefficient vector. The trend function ($f(\xi)$) can be a constant, linear and quadratic ([59]). However, in PCE-Kriging, the trend function is built on PCE orthonormal basis vector eq. (49). $Z(\xi, \omega)$ represents a stationary Gaussian process with zero mean and unit variance and σ^2 is the variance of Gaussian process [59].

3.4.3. Assessing model convergence

To evaluate the effectiveness of the developed model a metric such as adjusted R-squared [34] is considered as

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_{\text{model},i} - y_{\text{data},i})^2}{\sum_{i=1}^n (y_{\text{model},i} - \bar{y}_{\text{data},i})^2}. \quad (53)$$

3.5. Multi-objective robust design optimization of 5GDHC networks

The proposed framework performs ML-based robust optimization for 5GDHC networks. PCE-Kriging developed in section 3.4.2 is utilized as a ML model for the optimization. The overall objective of the formulated optimization problem is to optimize the components of financial metrics such as CAPEX_o, χ and TE. Heat demands, energy prices, installation costs, and CO₂ tax act as input uncertainties (ω_i), as shown in Fig. 7.

A systematic approach involves seeking a robust solution that performs well under the influence of uncertainties. Let $\omega_1, \omega_2, \dots, \omega_M$ be the probability distributions of the present uncertainties in the network. The robust nonlinear optimization problem is then can be defined as

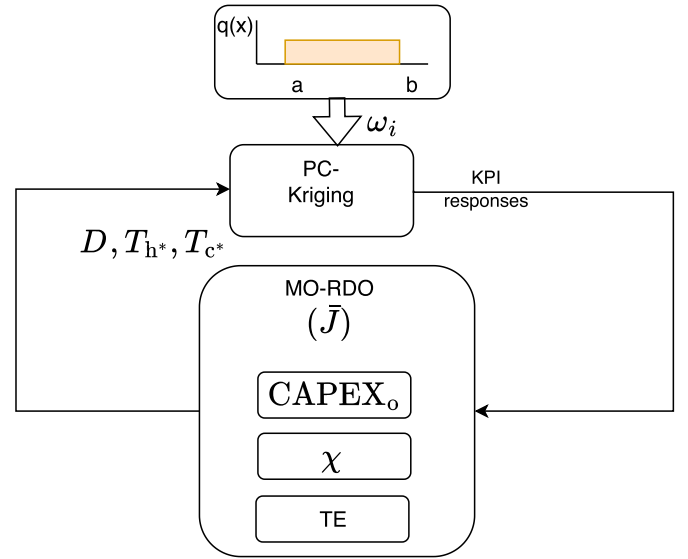


Fig. 7. Overview of PC-Kriging-based 5GDHC network optimization framework.

$$\min_{\forall Q_{\text{demand}}} \bar{J}(\omega_i) = \sum (\alpha \cdot \text{CAPEX}_o + (1 - \alpha) \cdot \text{TE}), \quad (54a)$$

$$\text{s.t. } D \in [D_{\min}, D_{\max}]^T, \quad (54b)$$

$$\mathcal{H} \in [\Delta p_{\min}, \Delta p_{\max}]^T, \quad (54c)$$

$$\Gamma \in [T_{c,\min}, T_{c,\max}]^T. \quad (54d)$$

The optimization routine selects the decision variable setting values by satisfying operational constraints as: eqs. (54b) and (54d) as:

$$80 \leq D \leq 250 \text{ mm},$$

$$11 \leq T_h \leq 35^\circ \text{C},$$

$$1 \leq T_c \leq 10^\circ \text{C},$$

$$50 \leq \mathcal{H} \leq 1000 \text{ Pa/m}.$$

$$(55)$$

In eq. (54a), the initial constraint on D is set to choose pipe diameters that are readily available in the DH market. The constraints on T_h and T_c delineate the limits for supply and return temperatures, aimed at operating the network in proximity to ambient and ground temperatures. The constraint on Δp ensures adequate mass flows and helps minimize peak power requirements for the pumps.

4. Case studies

Different locations from the Flanders region can be selected to apply a proposed framework for robust design analysis and optimization. But before introducing the particular case study, a small scale virtual network is considered to demonstrate the yearly-based KPIs and their impact on financial metrics.

4.1. Small virtual network

In Fig. 8 is shown that consists of TES, virtual building and a variable node.

The variable node is iterated with different types of buildings that have cooling and heating demands. For the virtual building node, an office with heating only and an office with both heating and cooling (prosumer type) are considered; see, for example, Table 10. The time series of energy demands for each node is generated using [43]. The yearly-based KPIs mentioned in Table 5 are evaluated using Fig. 5 considering the project life of 40 y and discount rate ($r = 4\%$).

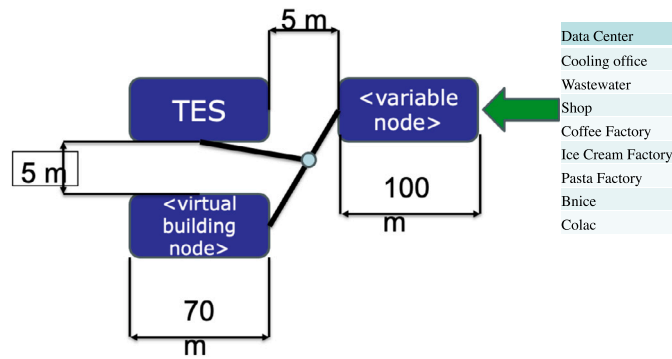


Fig. 8. Small virtual network with virtual office building node and a variable node of different type of prosumers.

Table 10

List of virtual scenarios with a small network of virtual building node and a variable node.

scenario 1		scenario 2	
variable node	building node	variable node	building node
Data Center		Data Center	
Cooling office		Cooling office	
Wastewater		Wastewater	
Shop		Shop	
Coffee Factory	office	Coffee Factory	cooling office
Ice Cream Factory		Ice Cream Factory	
Pasta Factory		Pasta Factory	
Bnice		Bnice	
Colac		Colac	

In Fig. 9, the correlation between the techno-economic and environmental KPIs listed in Table 5 is shown for two distinct scenarios, as outlined in Table 10. All KPIs are normalized to show the percentage improvements on a % scale. It is evident from Fig. 9 that approximately 20 % increase in both synergy and reverse flow duration (RV) leads to enhanced economic KPIs when considering a cooling-type building node iterated with a variable node. The spider web significantly expands in the case of a coffee factory paired with a cooling office, resulting in improvements across all KPIs.

On the financial assessment scale, NPV is evaluated for both scenarios as shown in Fig. 10a. It can be seen from Fig. 10a that NPV increases with a cooling office that acts as a prosumer in the network due to approximately 20 % improvements in yearly-based KPIs depicted in Fig. 9. Moreover, the saving-to-capex ratio also increases by employing a prosumer-cooling office in the network compared to the simple office building, as shown in Fig. 10b. Hence, all yearly-based KPIs influence the NPV calculated over a project's life.

4.2. Telenet 5GDHC network

For this study, a small location of Telenet from the Flander region is considered shown in Fig. 11.

This location consists of five buildings or nodes (Table 11), four of which are consumer types and one is a producer type, along with a centralized BTES. The time-series of energy demands for each node is generated using [43], and the uncertain demand variation is injected using eq. (1) with a specific surface area of the building SA and uncertain thermodynamic properties $EPC = 0.1$ to 0.4 MWh/m²y. Importantly producer node data center is assumed to have fixed energy supply over the season, thus no variations on it are considered, the demands of each node are listed in Table 12.

Each node is connected via pipe section with a specific diameter (D). As discussed in section 3.3, HDSS reads the layout plan of each node and section is exchanged with commercial tool **Zulu Thermo** which can also provide an initial guess of pipe sizes as listed in Table 9 against spe-

cific energy flows within the network. However, it does not estimate sizes under the influence of uncertain energy demands. Thus, based on available size ranges mentioned in [5], the uniform and bounded distribution of uncertainty in pipe sizes are considered which may result in uncertainties in associated pipe costs (eq. (7)).

As discussed in [8,26], the network's supply and return temperatures, as well as the heat pump's supply curves (S_h , S_c) to provide heating at desired temperature levels to prosumer premises, may vary due to differences in complementary load profiles. These variations are influenced by uncertainties in the overall energy demands, as outlined in Table 12. Therefore, uncertainty in the operational network temperatures based on the 5GDHC network studies [4,8,9] near ambient temperatures is considered and listed in Table 9.

Details of each node and diameter are listed in Tables 9 and 12.

The energy exchange of nodes with BTES over the season causes flow reversal in the network and classifies this configuration as a prosumer network [8].

5. Results and discussions

In this study, the results are divided into two main parts sections 5.1 and 5.2, followed by a discussion section 5.3. In section 5.1, a detailed analysis of KPI responses generated under multiple uncertain scenarios using a uniform distribution to develop the ML model is performed. In section 5.2, the developed ML model is used to identify influential uncertain design variables and objectives using the PCE approach. The PCE is further improved by the PCE-Kriging modeling approach to enhance the performance of the ML model by reducing the root mean squared error (RMSE) between the computational model HDSS and PCK. The ML model is then employed in MO-RDO section 5.2.3 to find the optimal trade-offs between environmentally optimal and economically optimal solutions to support the decision-making process for investors under the influence of uncertainties. The summary of the framework steps and associated outcomes are listed in Table 14 and the parameters related to energy markets are listed in Tables 7 and 8.

In the last section of the discussion section 5.3, first, we explained the significance of Sobol's at the network level, followed by comparing the MO-RDO solutions with non-optimal solutions, and their implications on financial metrics are analyzed.

5.1. Risk analysis via KPI responses

The results of risk analysis of financial assessment components such as NPV, IRR and DPBP for DC-Telenet against random variations are shown in Fig. 12. These components are calculated against KPI-response of CAPEX₀ and χ using eqs. (40) to (42). Whereas each of the response of KPI is calculated against uncertain random distribution of inputs within their uniform bounds as listed in Tables 8, 9 and 12. It can be seen from Fig. 12 that when the discount rate is varied between $r=0$ to 10%, the NPV remains greater than zero, with an IRR of 11 % and a DPBP of 11 y, even when all inputs are varied randomly. To identify the risk factors in the analysis, quantiles of NPV-PDF eq. (43) and CDF eq. (44) in the intervals of 10 %, 50 % and 90 % are estimated, respectively.

From Fig. 13, it can be seen that the PDF varies between the 10 % and 50 % quantiles, indicating a high variance of NPV against uncertain inputs. This shows that the variance of input uncertainties is propagated to the output NPV. Additionally, while the PDF in Fig. 13 provides insights into NPV outcome trends, it does not pinpoint the specific input variables influencing the NPV. Therefore, Sobol's analysis has been performed using PCE to identify the main uncertain inputs. The summary of the risk analysis before optimization is listed in Table 15.

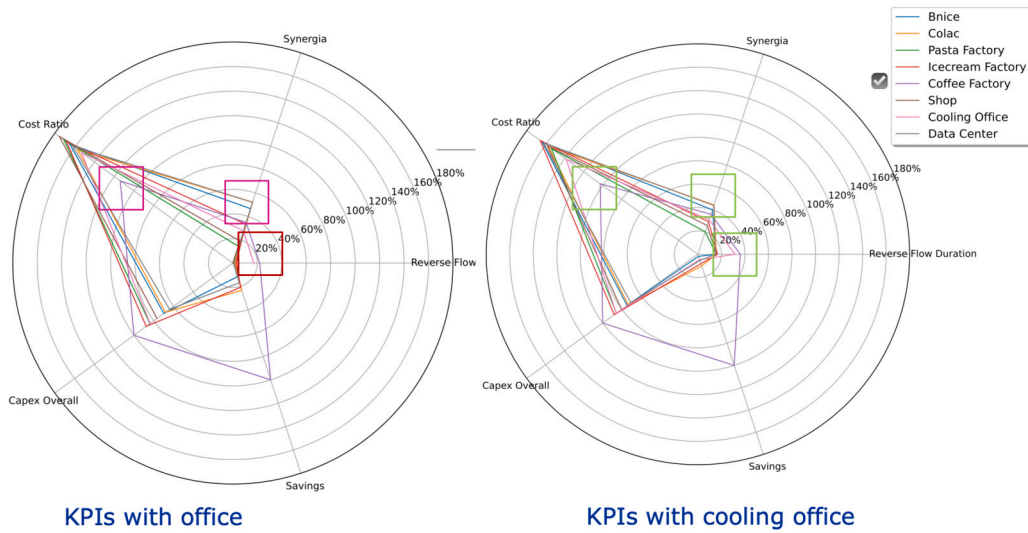


Fig. 9. KPIs comparison between two types of buildings with a variable source node. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

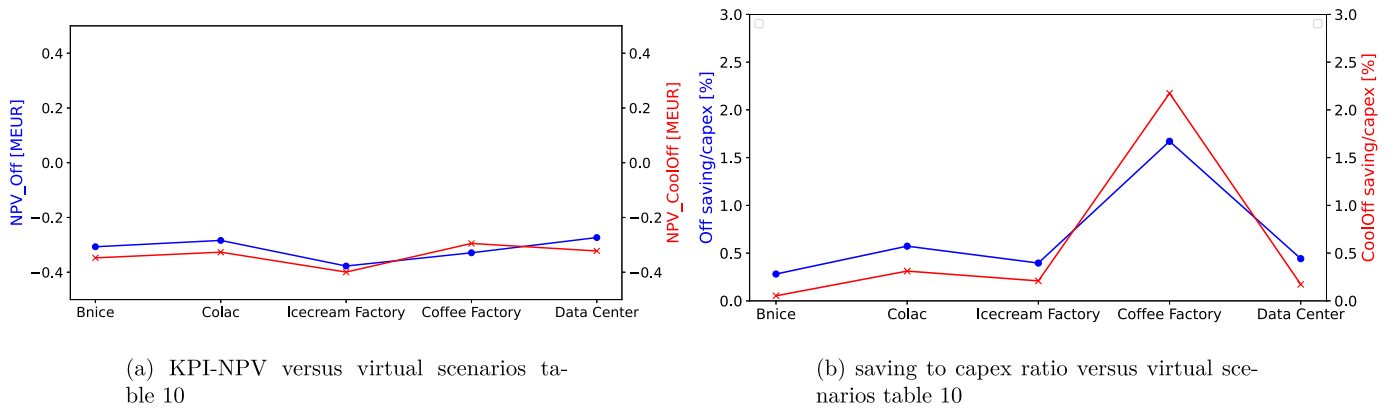


Fig. 10. NPV and savings-to-capex are shown against virtual network scenarios listed in Table 10.

Table 11

List of location from Flanders region of Belgium for the robust design analysis.

location	main producer	network length [m]	no. of buildings	Q_h [GWh]	Q_c [GWh]	synergy[%]
Telenet	data center	145	5	0.6	0.6	70

Table 12

Node types and energy demand variations of nodes present in DC-Telenet.

node	buildings	SA [m ²]	Q [MWh/y]	present type
N01	data center 1	237	608	SC
N02	office 2	596	60 to 238	SH
N03	office 6	5021	502 to 2008	SH
N04	office 8	375	37 to 150	SH
N05	office 10	743	74 to 297	SH

Table 13

Uncertainties in the network operational and design parameters.

description	symbol	initial	final	unit	source
pipe diameter	D_0	20	250	mm	market trend, [5]
	D_1	60	250	mm	
	D_2	40	20	mm	
	D_3	20	250	mm	
	D_4	20	250	mm	
	D_5	40	250	mm	
hot temperature	T_h	11	35	°C	[62,9]
cold temperature	T_c	1	10	°C	[62,9]
slope hot line	S_h	0.8	2.0	°C/°C	assumed
slope cold line	S_c	0.1	0.3	°C/°C	assumed

5.2. Machine learning results

5.2.1. Sobol's indices for DC-telenet

The proposed methodology relies significantly on employing PCE to calculate Sobol's indices (S_i). As previously stated, S_i plays a pivotal role in simplifying the complex Multi-Objective Robust Design Optimization (MO-RDO) by identifying appropriate decision variables, as illustrated in Fig. 1. The uncertain inputs considered of energy markets are based on Table 8, energy demand variations Table 12 and opera-

tional parameters Table 9, respectively. In total $d = 24$ uncertain inputs are considered for DC-Telenet. In order to improve the accuracy of computing S_i , 2000 samples are considered with a second-order polynomial ($p = 2$). The S_i for DC-Telenet are shown in Fig. 14.

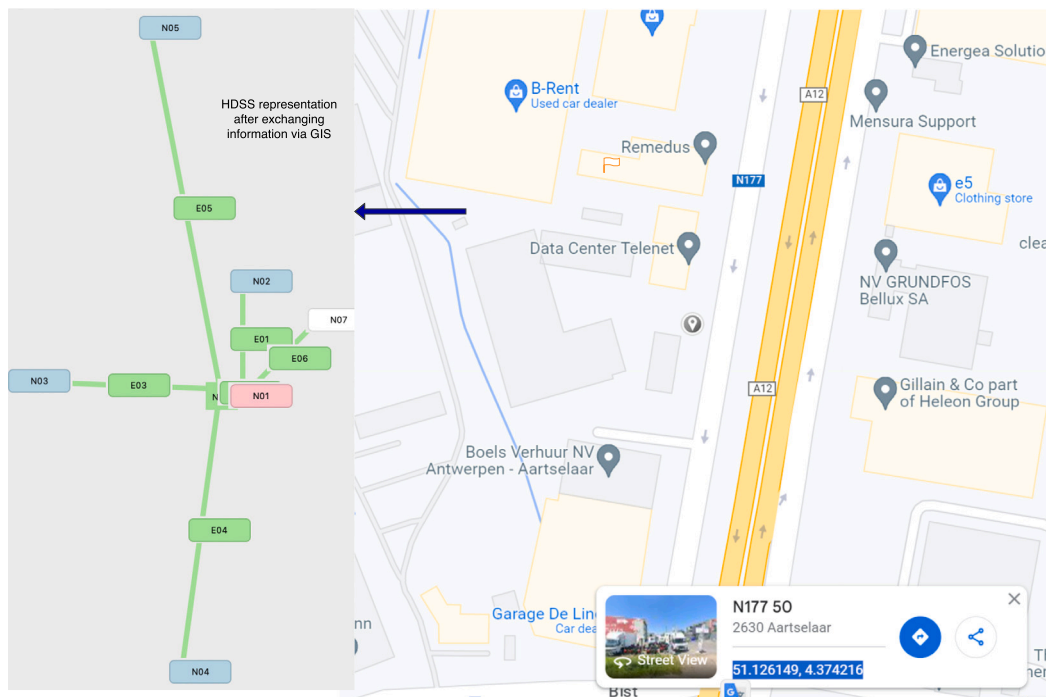


Fig. 11. Layout of Data Center Telenet is shown using HDSS simulator on left side, on rights side location is shown on google map.

Table 14

Summary of investigation steps used in this study for 5GDHC networks robust design and analysis.

step no.	description	inputs varied	outputs measured
1	risk analysis	uniform random variations using Tables 8, 12 and 13	network sizing and layout KPI responses(cpaex, savings, TE) financial metrics (NPV per meter, IRR and DPBP, PDF)
2	machine learning	KPI-responses PCE	PCE modeling Sobol's indices
3	multi-objective robust design and optimization (MO-RDO)	PCE Kriging design variables, constraints and objectives	design variables and objectives environmentally or economically efficient network designs

Table 15

Summary of the risk analysis of DC-Telenet.

description	KPI	value	unit	attractive location?
robust analysis	NPV	−2 to 4	kEUR/m	yes
	IRR	11	%	
	DPBP	11	y	
	PDF 10 to 50%	0.06 to 0.14	–	

From Fig. 14a, it can be seen that both network supply and return temperatures and investment costs together affect the overall investment of the network ($CAPEX_o$). The size of the storage and its investments are linked via network operating temperatures eq. (2). The network temperatures also influence other KPIs such as TE. This is because network temperature has a greater impact on the operations of decentralized and centralized substations. For instance, the COP of the heat pumps is directly influenced by network temperatures [8]. From Fig. 14b and Fig. 14c, it is apparent that for KPIs of net savings (χ) and total emissions (TE), uncertainties related to investment costs do not exert a significant influence. However, the impacts of operational-related inputs such as unit prices (c_{gas} , c_{el}), emission factor (e_{el}), and CO_2 tax, are notable. Regarding χ , c_{gas} has a higher impact than c_{el} even though the decentralized prosumer nodes are hydraulically coupled with the heat pumps. This is because χ is computed against boiler design technology as given in eq. (37). Thus, variations in c_{gas} significantly influence χ , and a low c_{gas} can even undermine the potential

business case. Moreover, due to the large surface area of Office 6 as indicated in Table 12 the uncertain demand variations of Office 6 which is significantly affect all KPIs.

The impact of uncertain input parameters related to energy market scenarios and building energy demands cannot be controlled, as discussed in [3]. For instance, energy demand fluctuations are influenced by evolving building characteristics. As a result, uncertain energy demands can lead to diverse network configurations and sizes [15]. Therefore, in this study, the adjacent diameter of the section pipe (D_2) is obtained to make the network robust against uncertain demand variations of Office 6. In Table 16, a summary of shortlisted uncertain inputs that interact with the KPIs is listed. As discussed above, not all uncertain inputs can be optimized. Therefore, the inputs that have $S_i > 0$ and can also act as design variables for the optimization problem eq. (54) to improve the KPIs are identified in Table 16. This will lead to reducing the design variables for the case of Telenet from 24 to only 3 (T_h , T_c and D_2).

Table 16

Key Input Parameters with High Sobol Indices and their importance in multi-objective robust design optimization (MO-RDO).

no.	uncertain input	$S_i > 0$ for KPI	MO-RDO	remarks
1	T_h and T_c	CAPEX _o	yes	control variable
1	Office 6	χ , CAPEX _o	no	from market
2	c_{el}	χ , TE	no	from market
3	c_{gas}	χ , TE	no	from market
4	c_{gas}	χ , TE	no	from market
5	c_{CO_2}	χ , TE	no	from market
6	D_2	CAPEX _o , χ and TE	yes	Office 6 demand variation

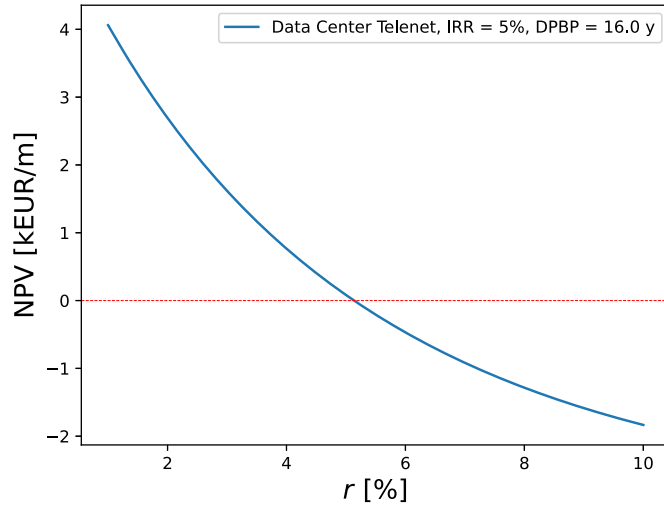


Fig. 12. NPV versus real discount rate (r), NPVs for every location is calculated with uniform random distribution of energy prices.

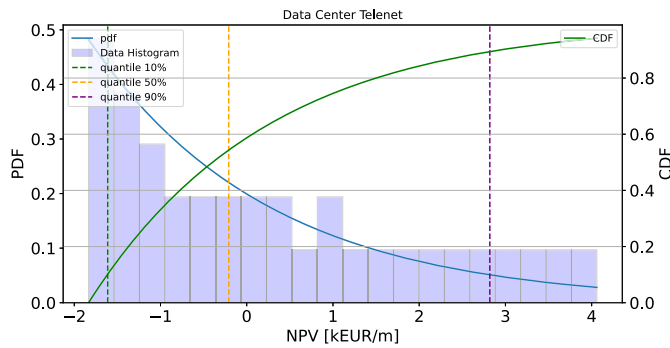


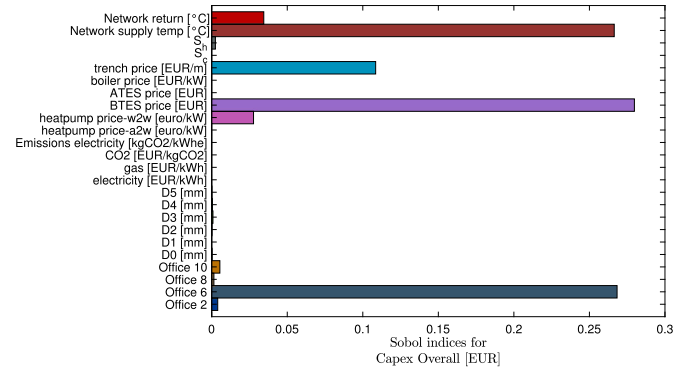
Fig. 13. NPV versus PDF and CDF.

5.2.2. PCK model performance

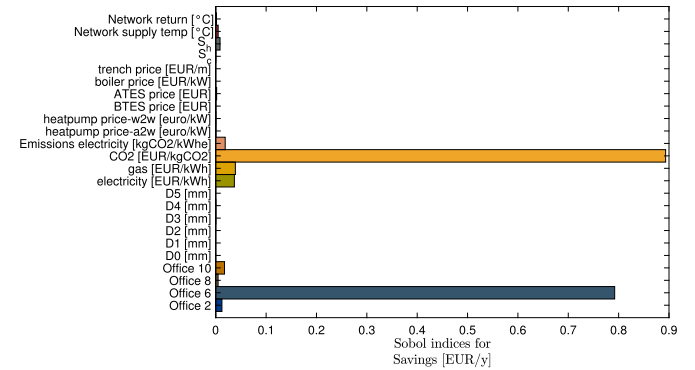
As discussed earlier, the HDSS will be replaced by an ML-based model in the optimization step to improve the computational efficiency. We considered the polynomial chaos-Kriging (PCK) approach as discussed in section 3.4.2. The model's performance is assessed using eq. (53). Analysis depicted in Fig. 15 reveals that most errors between the model output (y_{model}) and the actual data (y_{data}) lie along the linear line, indicating a significant level of accuracy of the ML-based model as discussed in [34].

5.2.3. MO-RDO

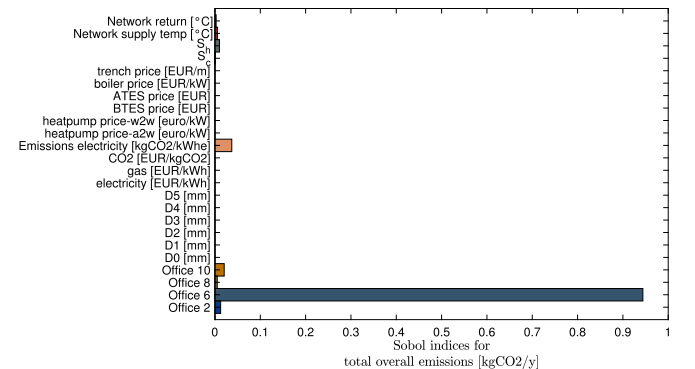
After the shortlisting of location and finding the decision variables and objectives via Sobol's the last step is to optimize the KPIs against uncertain inputs. The optimization problem formulated in eq. (54) is solved over the entire design space as depicted in Table 9, along with the influence of all uncertain inputs together.



(a) CAPEX_o against inputs



(b) Savings against inputs



(c) Total overall emissions

Fig. 14. Sobol' indices (S_i) for three KPIs are shown, $S_i \geq 0$ means the factor is influencing the output, $S_i \approx 0$ means the factor has no influence.

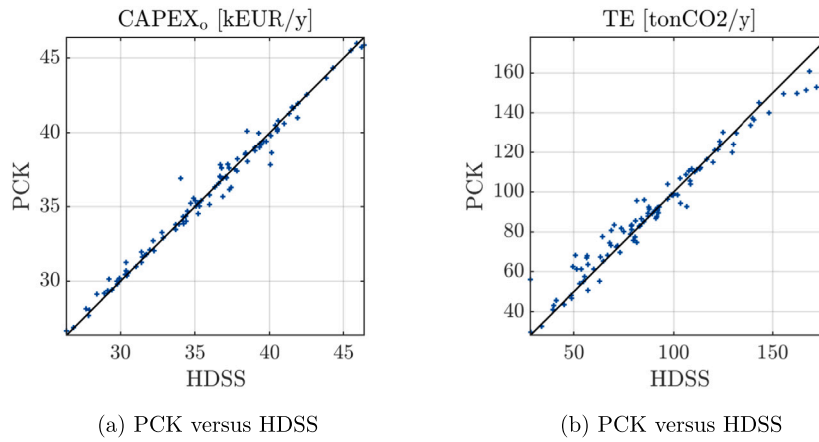


Fig. 15. Adjusted R-squared plot showing the performance of HDSS (y_{model}) and ML-model (y_{data}) for CAPEX_o and TE.

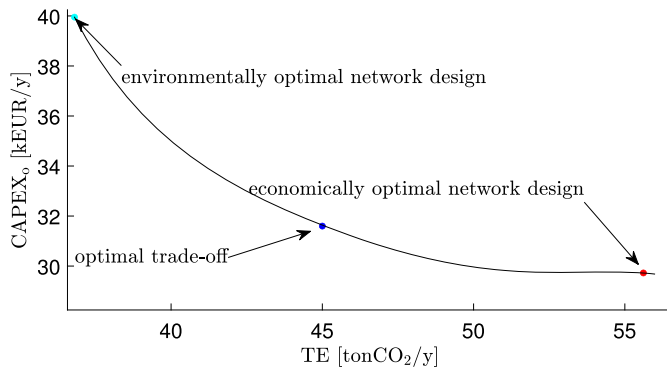


Fig. 16. Different design choices for CAPEX_o and TE.

The optimized solution in the formulated multi-objective problem provides a set of non-dominated designs (also called Pareto frontier points) of the 5GDHC network. These designs represent optimal trade-off solutions between the environmentally efficient and economically efficient objectives that satisfy the technical constraints eq. (54). It is important to mention that both qualitative and quantitative criteria are being used to evaluate design and objectives. First, the qualitative analysis is presented here in this section, and quantitative analysis is performed in section 5 with more insights and summarized in Table 18.

Qualitative analysis of MO-RDO: In this work, we have used MATLAB- `fmincon` nonlinear solver as it can be integrated with both equation-based model and data-driven models. The optimization eq. (54) yields a Pareto front between environmentally optimal and economically optimal solutions as shown in Fig. 16. The environmentally optimal solution corresponds to the design that has more network upfront cost but lower CO_2 emissions, whereas the economically optimal solution corresponds to lower network upfront investment cost but may result in higher CO_2 emissions. The optimal trade-off point is to guide investors in balancing between a more expensive but lower-emission solution and a cheaper option that may result in higher CO_2 emissions.

The evolution of the design variables is shown in Fig. 17 that lead to achieving objectives at distinct optimal points shown in Fig. 16. D_2 and ΔT are independent variables, as the network KPIs evolution differs with their variance as shown in Fig. 17a and 17b. In Fig. 17a, the evolution of D_2 is shown against two set objectives of CAPEX_o and TE. It can be seen from Fig. 17a that the optimization routine finds the high value of D_2 to get the environmentally optimal solution that results in a high value of CAPEX_o and results in reduced emissions of 35 tonCO_2/y . The other solution of economically efficient design can be obtained by selecting $D_2 = 120 \text{ mm}$. The exact trade-off can be seen where the evolution of two objectives intersects around $D_2 = 137 \text{ mm}$. In Fig. 17b,

the evolution of the objective is shown against the difference of supply and return temperatures (ΔT). The increase in ΔT will reduce the main cost component of CAPEX_o , such as storage cost, see for example in eq. (2). However, increasing ΔT will reduce the COP of decentralized heat pumps installed at prosumer premises, leading to higher electricity consumption and ultimately an increase in CO_2 emissions. In Table 17, a summary of the different MO-RDO designs is compared to the average mean of non-optimized solution.

5.3. Discussion

According to Sobol's indices, under uncertain energy markets, network temperatures are more important to consider when transitioning to 5GDHC networks. As such, a linear sensitivity analysis is performed using $T_c = 0, 5, 10, \dots, 35^\circ\text{C}$ and $T_h = 5$ to 35°C for network temperatures to observe effects on KPIs, as shown in Fig. 18. The AC of the form of eq. (34) and its decomposition are illustrated in Fig. 18a and Fig. 18b. Results show a nonlinear trend when T_h and T_c are varied. As shown in Fig. 18b the main component of CAPEX_o , such as capex storage, starts to decrease as ΔT increases between T_h and T_c . Variations in T_h and T_c have no significant impact on capex decentralized and capex pipes. A significant change occurs in OPEX_d and CO_2 cost decentralized as ΔT increases. The capex pipe, however, accounts for approximately 10 % of overall annualized expenses. Although we have considered D_2 due to high Sobol's of Office 6 demand variations, the main effect of different network solutions in terms of CAPEX_o and TE is determined by the best selection of network temperatures as indicated in Fig. 17b. χ and TE show opposite trends against T_h and T_c variations. The Fig. 18d illustrates that increasing ΔT will increase emissions, causing a rise in CO_2 emission costs and a reduction in χ as shown in Fig. 18c. Thus, a low combination of the network is favorable with centralized TES and to recover waste heat from the data center even though the initial upfront cost is high.

Quantitative analysis of MO-RDO: Both CAPEX_o and χ exert considerable influence on the NPV eq. (40). In Table 18, the main KPIs that contribute to the financial metrics are compared between MO-RDO solutions and non-optimized solutions. It can be seen from Table 18 that the economically optimal solution of MO-RDO is close to the non-optimized solution obtained from HDSS. As discussed in section 3.3 HDSS sizes the network assets based on peak demands and does not account for uncertainties influence in the asset size optimization. By selecting optimization weight $\alpha > 0.5$ in eq. (54), the MO-RDO prioritizes the network investment like the HDSS. However, choosing $\alpha < 0.5$ will lead to the environmental solution, and will increase network investment costs by 10 %, while reducing CO_2 emissions by 13 %, ultimately causing χ to increase. In terms of financial metrics to evaluate the project feasibility over its life, the NPV of environmentally optimal solutions increases

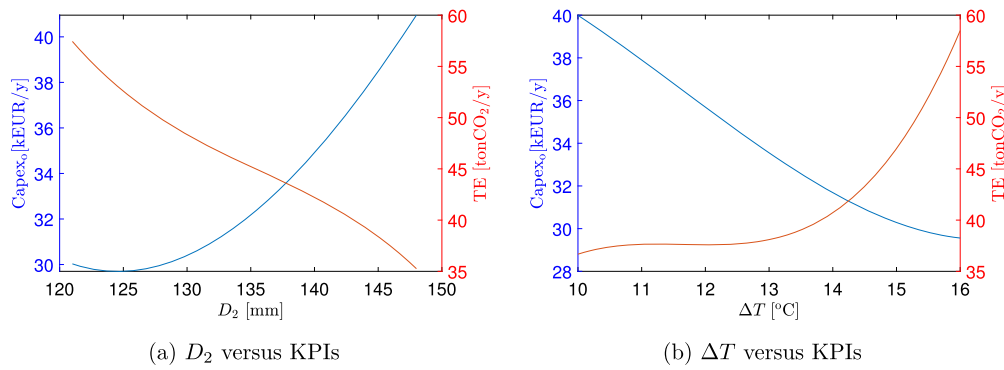


Fig. 17. Optimal network parameters to yield optimal environmental and economical performances.

Table 17

Summary of robust optimal designs for DC-Telenet.

description	symbol	non-optimized mean values	MO-RDO environmentally optimal	MO-RDO optimal-trade-off	MO-RDO economically optimal	unit
supply temperature	T_h	20	15	20	22	°C
return temperature	T_c	6	5	6	5	°C
selected section	D_2	102	150	135	125	mm

Table 18

Summary of KPI comparison for DC-Telenet.

description	symbol	non-optimized mean values	MO-RDO environmentally optimal	MO-RDO economically optimal	unit
overall capex	CAPEX ₀	815	887	812	kEUR
net savings	χ	80	83	80	kEUR/y
total emissions	TE	21	18	20	tonCO ₂ /y
net present value	NPV	4.0	3.8	4.0	kEUR/m
discounted payback period	DPBP	16	17	16	y

by only 3 % compared to economically efficient solutions that are not optimized. In addition, the DPBP increases only by 1 y.

6. Conclusion

In this study, a robust design and optimization framework was presented to optimize the design parameters of a 5GDHC network under the influence of uncertain energy markets and the energy supplies and demands of the participating prosumers. 2000 uncertain scenarios are generated for a potential 5GDHC business case, and an ML is employed on a uniformly distributed input and output data set. The performance of the ML model is analyzed in comparison with the data set, which has been generated from a Python-based high-fidelity model. In general, almost 90 % linear agreement between the high-fidelity model and ML model holds for CAPEX₀ and TE. As a result, the ML model replaces the computationally inefficient model of the 5GDHC network with a simple surrogate. Before solving the complex formulated multi-objective robust design optimization problem (MO-RDO), the proposed ML model identifies critical design variables in the presence of numerous uncertain inputs from energy markets and the uncertain thermal properties of participants. This has been accomplished by performing global sensitivity analysis, yielding Sobol's indices to quantify the impact of each input on output.

For the considered case study, high Sobol's indices for gas prices indicate a greater impact than electricity prices, suggesting that lower gas prices could undermine the 5GDHC business case. Additionally, Sobol's analysis identifies which buildings most significantly affect the network KPIs due to variations in demand and thermal characteristics. Accord-

ing to Sobol's analysis, only 6 from the considered 24 inputs have a high impact. At the same time, only 3 of the 6 high Sobol's inputs are considered for the MO-RDO, being the supply (T_h) and return (T_c) temperature of the network and the diameter of the network pipes to the building with high Sobol's. Consequently, the proposed methodology reduced the computational complexity of the optimization problem by systematically reducing the number of design variables for large networks.

The overall objective of the proposed methodology is to maximize the financial metrics such as NPV. The components of NPV, as given in eq. (40), are optimized along with the objective to reduce the TE in the presence of uncertainties. This transforms the network optimization problem into MO-RDO of the form eq. (54). The optimization framework selects the best combination of decision variables that can optimize the objectives as given in eq. (54). Each combination represents a variety of optimal designs (environmentally optimal or economically optimal) and trade-offs that form a Pareto frontier. In contrast to the economically optimal design, the environmental design had a 10 % higher investment cost but reduced TE by 13 % and improved net savings. Only a 3 % increase in NPV is estimated as a risk for the environmentally optimal solution on a financial scale. Thus, following the EU energy policy for CO₂ reduction, the environmental design anticipates only a 3 % reduction in NPV over the project life, despite higher investment costs and uncertain energy markets.

The methodology enhances flexibility in identifying key uncertain input parameters and evaluating the robustness of critical performance indicators, addressing the limitations of current GIS-based approaches. Additionally, applying MO-RDO to the uncertain dataset improves

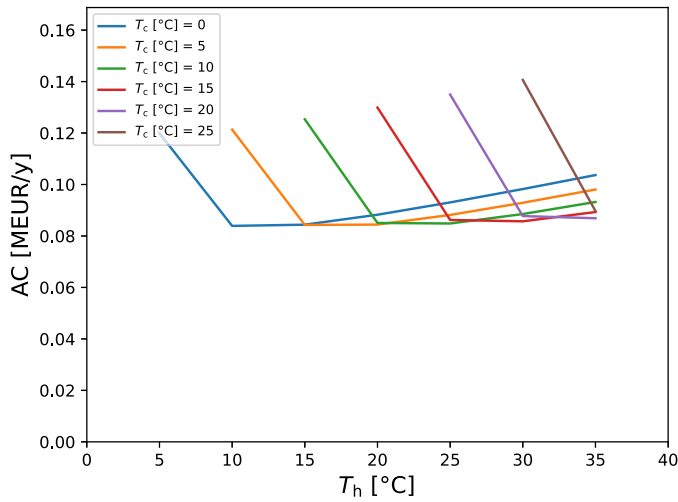
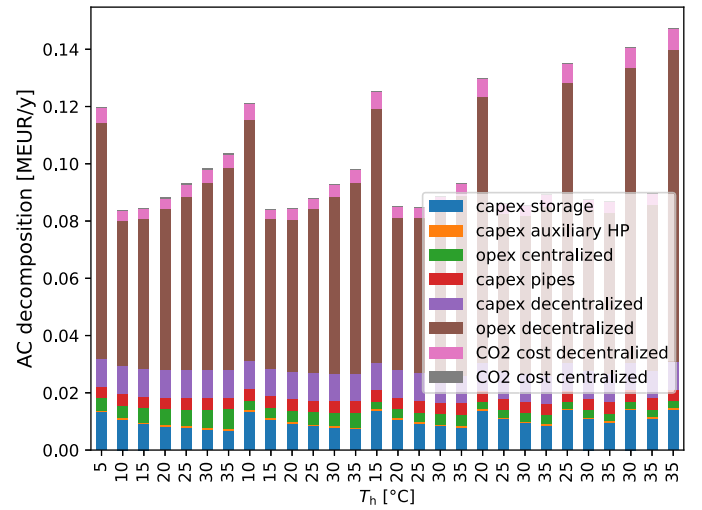
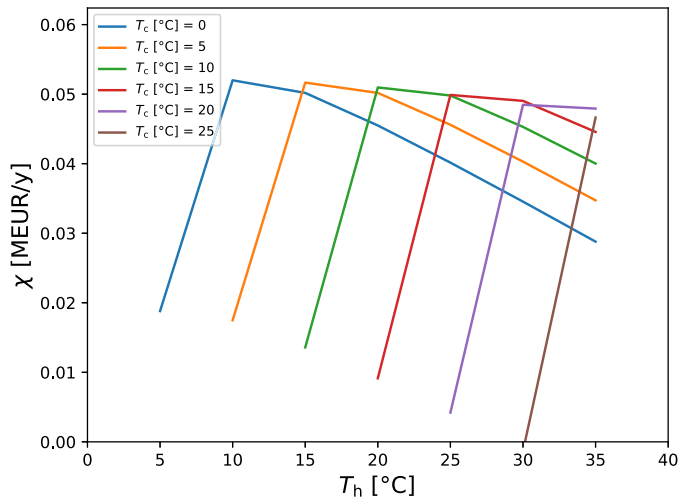
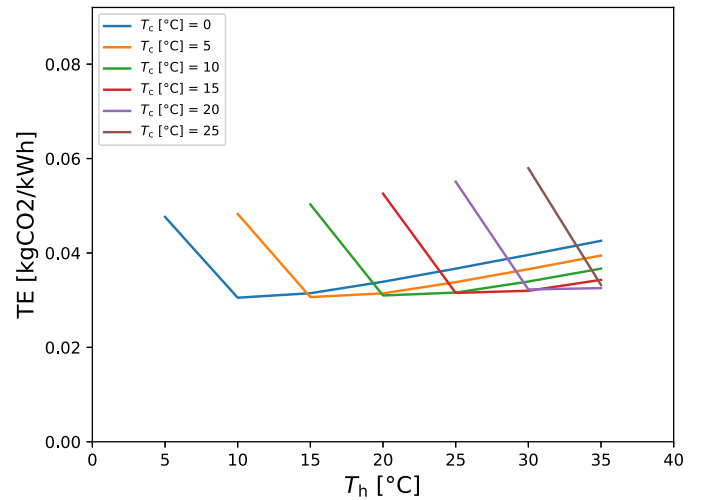
(a) AC against T_h and T_c (b) AC decomposition against T_h and T_c (c) χ decomposition against T_h and T_c (d) TE decomposition against T_h and T_c

Fig. 18. DC-Telenet AC and its decomposition is shown against unique cold-side network temperature $T_c = 0, 5, 10, \dots, 35^\circ\text{C}$, and varying hot-side temperature $T_h = 5$ to 35°C .

decision-making support and serves as a step towards uncertainty-based robust optimization for 5GDHC networks. However, the approach falls short in online optimization, as it lacks the ability to generate random scenarios and optimize simultaneously. Future work will incorporate the robust design optimization methodology from the Python framework RHEIA [63], which employs the Non-dominated Sorting Genetic Algorithm (NSGA-II) combined with Polynomial Chaos Expansion (PCE) to explore trade-offs between economic and environmental objectives under uncertainty.

CRediT authorship contribution statement

Afraz Mehmood Chaudhry: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Ghader Ghorbaniasl:** Writing – review & editing, Validation, Supervision, Investigation, Formal analysis, Data curation. **Jonathan Hachez:** Visualization, Software, Resources, Formal analysis, Data curation, Conceptualization. **Stanislav Chicherin:** Writing – review & editing, Visualization, Validation, Re-

sources, Formal analysis, Data curation. **Svend Bram:** Writing – review & editing, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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